CeNTREX State Cleanup

Design and optical-Bloch-equation simulations of the rotational-state cleanup stages CU1 ($J = 1$) and CU2 ($J = 2$)

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Abstract

The CeNTREX experiment measures the Schiff moment of the ^{205}Tl nucleus in a beam of thallium fluoride. After the electrostatic lens the molecules are in $X(v{=}0)$, $J = 2$, and state preparation (SPB) transfers them to $J = 1$, the level used in the interaction region. Two state-cleanup stages bracket this transfer: CU1 removes the population of $J = 1$ before SPB, and CU2 removes the population left in $J = 2$ after it. Both stages pump the molecules through the $B\,^3\Pi_1$ state into other rotational levels, in the electric field that is kept on to preserve the spin. In that field the Stark interaction spreads the excited-state sublevels over tens of MHz, so that a single laser frequency leaves many ground-state superpositions dark. This report documents how the two cleanup schemes were designed around this problem, and simulates both with Lindblad master equations using `centrex-tlf` 0.2.5 and its native Rust solver.

The model contains the complete field-dressed hyperfine structure of the addressed levels, including both Λ-doublet components of B; decay into every rotational level of X with branching ratios computed in the field; a 13-pass multipass beam with 10 % loss per pass; X/Z polarization switching at 0.5 MHz; and phase modulation. A single molecule at 184 m/s is followed through 5 cm. Its transverse velocity (with the Doppler shift that alternates in sign from pass to pass) and transverse position are scanned as well, and the results are averaged over the transverse distribution of a beam simulation (lens at 25 kV), counting only molecules inside the detection laser.

CU1 uses one 80 mW laser on R(1) $F_1' = 5/2$, $F' = 2$ at 150 V/cm, without phase modulation. On the beam axis it removes 99.96 % of the population at +20.2 MHz from the main resonance. The optimum is narrow: the cleanup exceeds 99.9 % only between +20.0 and +20.5 MHz. It is therefore very sensitive to the transverse velocity: 99.8 % at 0.3 m/s, 97.4 % at 0.5 m/s and 78 % at 1 m/s. For a Gaussian velocity distribution of 1.8 m/s FWHM (truncated at 1 m/s) and a uniform transverse position within 15 mm, the average cleanup is 93.1 % at +20.2 MHz and 95.2 % at +19.0 MHz, the better detuning for this distribution. A 1.6 MHz phase EOM ($\beta = 1.5$) raises it to 96.2 %. The gain is small, so the EOM is not adopted for now and +19.0 MHz is taken as the CU1 set point. Under the present conditions, a 3 cm high beam and the beam-simulation distribution ($\sigma_{v_x} = 0.65$ m/s, $\sigma_y = 11$ mm) within the detection acceptance ($|v_x| \le 1$ m/s, $|y_0| \le 8.5$ mm), +19.0 MHz is again the best detuning and CU1 removes 98.7 % of the $J = 1$ population. At this detuning the molecules at rest (99.0 %) are cleaned less well than those at $|v_x| \approx 0.5$ m/s (99.99 %), whose Doppler shift brings the SPB-related sublevels into resonance in half of the passes.

CU2 drives R(2) $F_1' = 7/2$, $F' = 3$ (80 mW) and $F' = 4$ (24 mW) with two beams derived from one laser by an AOM (200 ± 15 MHz), both phase-modulated at 17.5 MHz with $\beta = 1.5$, in a 2.4 cm high beam. The experiment now operates CU2 at about 200 V/cm. There the $F' = 4$ main resonance lies 211.83 MHz above the $F' = 3$ main resonance, compared

with 219.93 MHz at the original 250 V/cm. The detuning pair found at 250 V/cm (99.99 % there) therefore gives only 92.9 % at 200 V/cm. A two-dimensional optimisation of laser and AOM frequency over the full AOM range finds 99.85 % at +11.5 MHz from the $F' = 3$ main resonance and $f_{\mathrm{AOM}} = 191.0$ MHz, confirmed with tighter tolerances. A nearly equivalent optimum, 99.80 %, lies at $f_{\mathrm{AOM}} = 208.5$ MHz. Both are 2–3 MHz wide at the 99 % level. With the present 3 cm beam and the transverse distribution within the detection acceptance, the second optimum is the more robust one: at $f_{\mathrm{AOM}} = 208.5$ MHz CU2 removes 96.7 % of a uniformly populated $J = 2$ level and 99.6 % of the population left in the SPB source levels. For CU2 the transverse position matters more than the transverse velocity.

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Chapter 1

Introduction

1.1 CeNTREX

CeNTREX (Cold molecule Nuclear Time-Reversal EXperiment) [1] searches for the Schiff moment of the ^{205}Tl nucleus, a CP-violating nuclear moment, by measuring a CP-odd frequency shift in thallium fluoride (TlF). It uses a cryogenic buffer-gas beam of TlF, rotational cooling, an electrostatic lens, state preparation, a Ramsey measurement in a strong electric field and laser-induced-fluorescence detection. Every step relies on the molecules being in well-defined rotational and hyperfine levels of the ground electronic state.

1.2 State cleanup in the CeNTREX beamline

The molecules leave the electrostatic lens in $X(v{=}0)$, $J = 2$. The state-preparation stage SPB then transfers them from $J = 2$ to $J = 1$, the rotational level used in the interaction region. Two cleanup stages bracket this transfer (Fig. 1.1):

- **CU1**, before SPB, pumps off the population of $J = 1$. Molecules already in $J = 1$ would otherwise reach the interaction region together with the transferred ones.
- **CU2**, after SPB, pumps off the population that remains in $J = 2$.

Ideally the cleanup would also remove the other sublevels of $J = 2$ before SPB and of $J = 1$ after it, except the target state. That would require a broadband UV EOM and is not pursued.

A cleanup stage removes the population of a rotational level by optical pumping. The molecules cross a laser resonant with a transition from that level to $B\,^3\Pi_1$. Each excitation is followed by spontaneous decay, and because B decays to several rotational levels of X, most decays end in a level that the light does not address. The remaining decays return the molecule to the addressed level, where it can be excited again. A few photons per molecule therefore suffice, provided that every sublevel of the addressed level is excited and none is left in a dark superposition.

To preserve the spin, the electric field is kept non-zero until the end of the beamline, so both stages pump in an electric field: about 150 V/cm for CU1 and, in the present configuration, about 200 V/cm for CU2 (250 V/cm in the original design). In the field the Stark interaction moves the hyperfine sublevels of the excited state apart by tens of MHz, many natural linewidths. A single laser frequency then reaches only a few of them, and more ground-state superpositions are dark. The design of both schemes (Chapter 3) is mainly a

response to this problem.

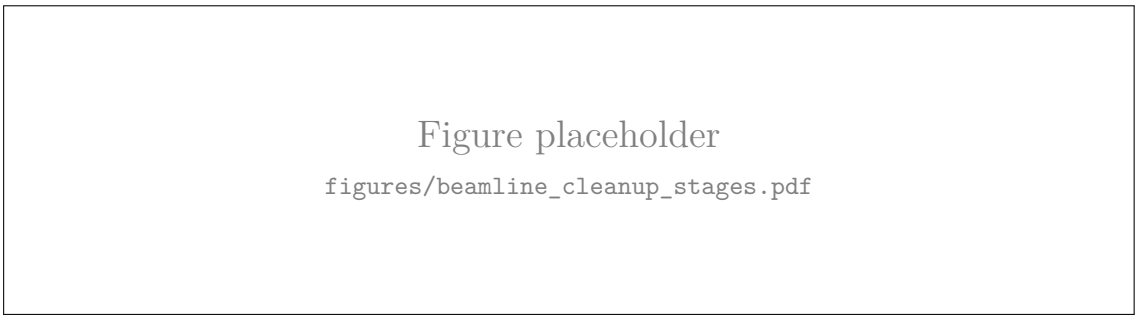


Figure 1.1: Cleanup stages in the CeNTREX beamline: lens ($J = 2$) → CU1 (removes $J = 1$) → SPB ($J = 2 \to J = 1$) → CU2 (removes $J = 2$) → interaction region ($J = 1$). Design-study slide 2 [2].

1.3 Scope of this work

The aims are

1. to document the design of the two cleanup schemes;
2. to set up a complete optical-Bloch model of each stage, with the field-dressed level structure, the multipass beam geometry, polarization switching, phase modulation and, for CU2, the AOM coupling of the two beams;
3. to find the laser frequencies that maximise the cleanup efficiency within the hardware limits, and the width of the optimum;
4. to establish which approximations the results depend on, and to check the numerics.

All results use the baseline parameters of Table 1.1. The model is a single-trajectory model: one molecule at the mean forward velocity. The transverse velocity, with its Doppler shift, and the transverse position are scanned, and the results are averaged over the transverse distribution of a beam simulation within the detection acceptance (CU1: Secs. 5.6–5.10; CU2: Sec. 6.7). **Two beam heights appear in this report.** The design-level results (detuning, power and the two-dimensional CU2 optimisation) use 4.8 cm for CU1 and 2.4 cm for CU2. In the experiment both beams are now about 3 cm high; the results for these present conditions are in Secs. 5.10 and 6.7.

1.4 Outline

- Chapter 2 introduces the TlF levels, the cleanup transitions and their structure in the applied field.
- Chapter 3 describes how the two schemes were designed, and the operating constraints, in particular the AOM coupling of the CU2 beams.
- Chapter 4 describes the simulation in detail and summarises every assumption.
- Chapters 5 and 6 present the CU1 results and the CU2 results at 200 V/cm.

Table 1.1: Baseline parameters of this report. Beam heights are full 4σ widths.

	CU1 ($J = 1$)	CU2 ($J = 2$)
electric field	150 V/cm	200 V/cm
transition(s)	R(1) $F_1' = 5/2$, $F' = 2$	R(2) $F_1' = 7/2$, $F' = 3$ and $F' = 4$
laser power	80 mW	80 mW ($F' = 3$) + 24 mW ($F' = 4$)
beam height	4.8 cm; present: 3 cm (Sec. 5.10)	2.4 cm; present: 3 cm (Sec. 6.7)
detuning	+20.2 MHz on axis; set point +19.0 MHz	optima R1 and R2 (Sec. 6.6)
phase modulation	none	17.5 MHz, $\beta = 1.5$, common to both beams
AOM	–	$f_{\mathrm{AOM}} = 200 \pm 15$ MHz
multipass	13 passes, 10 % loss per pass, 3 mm wide along the molecular beam	
polarization	X/Z switching at 0.5 MHz	
molecule	184 m/s on the beam axis, 5 cm interaction length	
transverse motion	v_x and y_0 scanned (Secs. 5.6, 5.10)	v_x and y_0 scanned at the optima (Sec. 6.7)
transverse distribution	beam sim. (25 kV): $\sigma_{v_x} = 0.65$ m/s, $\sigma_y = 10.95$ mm	beam sim. (25 kV): $\sigma_{v_x} = 0.65$ m/s, $\sigma_y = 10.35$ mm
detection acceptance	$\lvert v_x \rvert \leq 1$ m/s, $\lvert y_0 \rvert \leq 8.5$ mm (detection laser 1.7 cm high)	

- Chapter 7 collects the numerical checks, and Chapter 8 concludes. The appendices give the full parameter list, the 250 V/cm reference results and the code and data structure.

Chapter 2

Physical system

2.1 The TlF states

The ground electronic state $X\,{}^1\Sigma^+$ ($\Omega = 0$) and the excited state $B\,{}^3\Pi_1$ ($\Omega = \pm 1$) both carry hyperfine structure from the two nuclear spins, $I_{\mathrm{Tl}} = I_{\mathrm{F}} = 1/2$. Following the `centrex-tlf` conventions [3], the spins couple to the rotation in two steps, $\mathbf{J}+\mathbf{I}_1 = \mathbf{F}_1$ and $\mathbf{F}_1+\mathbf{I}_2 = \mathbf{F}$, and states are labelled $|J, F_1, F, m_F, P\rangle$. Each rotational level of X has a single parity $P = (-1)^J$. Each rotational level J' of B is a Λ-doublet with one component of each parity. The Λ-doubling is small, so the two components are strongly mixed by electric fields of a few hundred V/cm. Primed quantum numbers refer to B. A transition is named by its branch and the ground-state J: R(J) is $J \to J' = J + 1$.

At zero field the $J = 1$ level has 12 sublevels ($F_1 = 1/2$: $F = 0, 1$; $F_1 = 3/2$: $F = 1, 2$), and $J = 2$ has 20 ($F_1 = 3/2$: $F = 1, 2$; $F_1 = 5/2$: $F = 2, 3$).

2.2 The cleanup transitions

Table 2.1 lists the transitions and the *main* state pairs. The main pair of a laser fixes the zero of its detuning axis and normalises its Rabi rate (Sec. 4.5). Its matrix element m, in units of the reduced X–B transition dipole moment, is evaluated between the field-dressed states for X polarization.

Table 2.1: Cleanup transitions, main state pairs and main-pair matrix elements m (X polarization, field-dressed states).

stage	transition	main pair $\lvert J, F_1, F, m_F\rangle \to \lvert J', F_1', F', m_F', P'\rangle$	$\lvert m\rvert$
CU1, 150 V/cm	R(1) $F_1' = 5/2$, $F' = 2$	$\lvert 1, \frac{3}{2}, 2, 1\rangle \to \lvert 2, \frac{5}{2}, 2, 2, +\rangle$	0.1004
CU2, 200 V/cm, beam 0	R(2) $F_1' = 7/2$, $F' = 3$	$\lvert 2, \frac{5}{2}, 3, 2\rangle \to \lvert 3, \frac{7}{2}, 3, 3, -\rangle$	0.0948
CU2, 200 V/cm, beam 1	R(2) $F_1' = 7/2$, $F' = 4$	$\lvert 2, \frac{5}{2}, 3, 0\rangle \to \lvert 3, \frac{7}{2}, 4, 1, -\rangle$	0.2286

2.3 Structure in the applied field

The cleanup takes place in a static electric field $\mathbf{E} = E\hat{\mathbf{z}}$, which defines the quantisation axis. With $\mathbf{E}$ and the small magnetic field both along z, m_F remains a good quantum number. J, F_1, F and, in B, the parity are mixed. In the ground state the hyperfine structure of each rotational level spans only a few MHz (2.4 MHz for $J = 1$ at 150 V/cm, 1.4 MHz for $J = 2$ at 250 V/cm). In the excited state the $|m_F'|$ sublevels of one hyperfine level are split

by the Stark interaction by about 10–25 MHz each (Chapter 3), much more than the natural linewidth $\Gamma/2\pi = 1.61\,\mathrm{MHz}$.

Table 2.2 gives the positions of all transition components with a non-negligible matrix element, measured from the main resonance of the beam that drives them. These positions set the optimum detunings and are the reason for phase modulation in CU2.

Table 2.2: Spread of the transition components (pairs with $|M_{eg}|^2 > 10^{-4}\max|M_{eg}|^2$, X and Z polarization combined) relative to the main resonance of the beam that drives them, and their mean weighted by $|M_{eg}|^2$. Computed from the rotating-frame Hamiltonian of the OBE system.

system	pairs	lowest	highest	weighted mean
CU1, 150 V/cm, addressed parity only	32	0.0 MHz	34.2 MHz	19.5 MHz
CU1, 150 V/cm, both parities	62	0.0 MHz	79.9 MHz	33.3 MHz
CU2, 200 V/cm, beam 0 ($F' = 3$)	106	0.0 MHz	86.9 MHz	40.7 MHz
CU2, 200 V/cm, beam 1 ($F' = 4$)	106	−31.4 MHz	55.5 MHz	8.9 MHz

For CU2 the two beams address different excited hyperfine levels. Their main resonances are separated by

$$\Delta_{\mathrm{main}}(E) = \frac{(E_{e_0} - E_{g_0})_1 - (E_{e_0} - E_{g_0})_0}{2\pi} = \begin{cases} 211.83\,\mathrm{MHz}, & E = 200\,\mathrm{V/cm}, \\ 219.93\,\mathrm{MHz}, & E = 250\,\mathrm{V/cm}, \end{cases} \tag{2.1}$$

with the $F' = 4$ resonance above the $F' = 3$ one. The physical frequency difference of the two beams is therefore

$$\frac{\omega_1 - \omega_0}{2\pi} = \Delta_{\mathrm{main}}(E) + \frac{\delta_1 - \delta_0}{2\pi}, \tag{2.2}$$

with δ_k the detuning of beam k from its own main resonance. In the experiment this difference is the AOM frequency (Sec. 3.5.1).

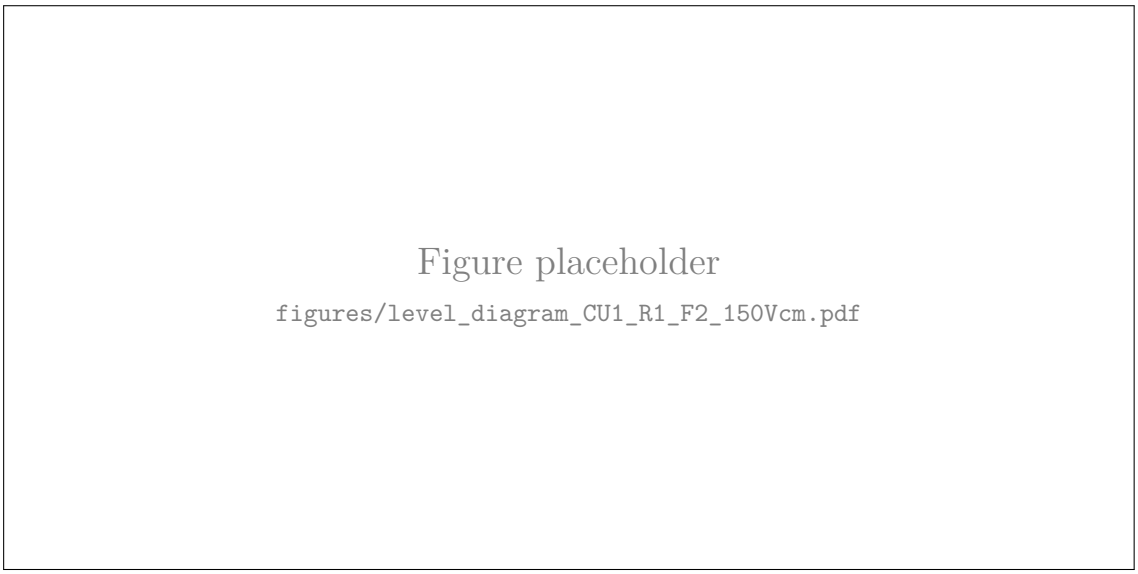


Figure 2.1: Field-dressed $X \to B$ level diagram for R(1) $F_1' = 5/2$, $F' = 2$ at 150 V/cm, with each level segmented by its zero-field parent character (to be generated with `utils.plotting.plot_transition_level_diagram`).

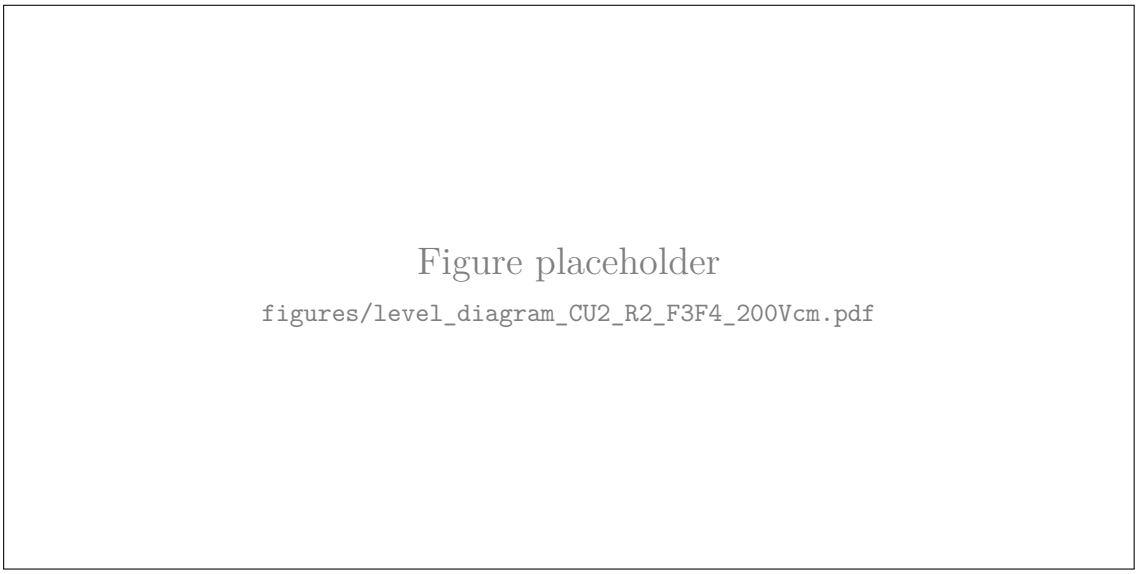


Figure 2.2: As Fig. 2.1 for R(2) $F_1' = 7/2$, $F' = 3$ and $F' = 4$ at 200 V/cm.

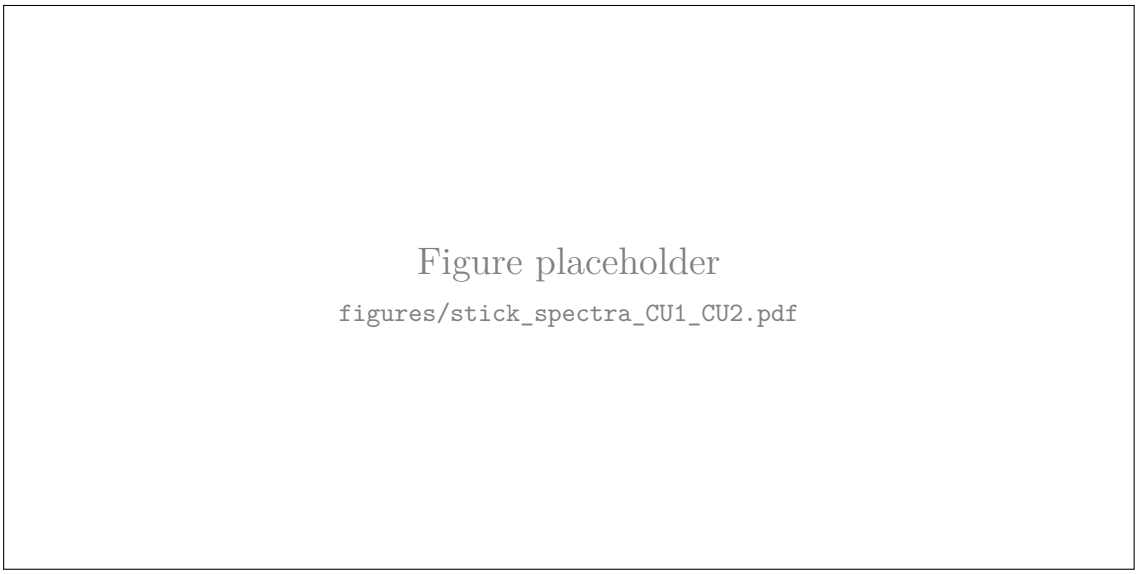


Figure 2.3: Stick spectra of the transition components of Table 2.2: position on each beam's detuning axis and $|M_{eg}|^2$ relative to the main pair. (a) CU1 at 150 V/cm. (b) CU2 at 200 V/cm, beams 0 and 1, with the carriers and their phase-modulation sidebands.

2.4 Field mixing of the ground-state hyperfine levels

The field also mixes the ground-state hyperfine levels, which matters for the cleanup because it opens transitions that are forbidden at zero field. The clearest case is $|J = 1, F_1 = \frac{1}{2}, F = 0, m_F = 0\rangle$. At zero field it cannot be excited to $F' = 2$ ($\Delta F = 2$). In the field it acquires an admixture of $|J = 1, F_1 = \frac{3}{2}, F = 2, m_F = 0\rangle$ (Table 2.3), which at 150 V/cm carries 29 % of the probability, and this admixture is what CU1 pumps. The same mechanism lets CU2 pump the $F = 1$ sublevels of $J = 2$ to $F' = 3$, through their admixture of $F = 3$ (Sec. 3.3).

Table 2.3: Composition of the field-dressed state $|J = 1, F_1 = \frac{1}{2}, F = 0, m_F = 0\rangle$ (amplitudes) versus electric field (design study [2], slide 9).

E	$\lvert F_1 = \frac{1}{2}, F = 0, m_F = 0\rangle$	$\lvert F_1 = \frac{3}{2}, F = 2, m_F = 0\rangle$
0 V/cm	1.00	–
20 V/cm	1.00	0.08
40 V/cm	0.97	0.26
150 V/cm	0.84	0.54

2.5 Branching ratios

Table 2.4 gives the decay branching ratios of the addressed excited levels into the rotational levels of X, computed between the field-dressed states. They show why a few photons suffice: from $F' = 2$ of B, $J' = 2$ only 37 % of the decays return to $J = 1$, and from $F' = 3$ or $F' = 4$ of $J' = 3$ only 27 % return to $J = 2$. The field-dressed B levels contain a large opposite-parity admixture. Even when only the addressed parity is kept as a dynamical level (CU1 row), 28 % of the decays go to $J = 2$, which is parity-forbidden at zero field.

Table 2.4: Branching ratios of the addressed B levels into the rotational levels of $X(v{=}0)$ in the applied field. For the CU2 levels both Λ-doublet components are retained and the ratios are summed over all excited sublevels of the level and normalised.

excited level	$J = 0$	$J = 1$	$J = 2$	$J = 3$	$J = 4$	$J = 5$
$J' = 2$, $F' = 2$, $P' = +$ (150 V/cm)	0.00005	**0.371**	0.280	0.334	0.016	–
$J' = 3$, $F' = 3$ (200 V/cm)	–	0.00002	**0.265**	0.484	0.235	0.016
$J' = 3$, $F' = 4$ (200 V/cm)	–	0.00001	**0.265**	0.484	0.235	0.016

Chapter 3

Design of the cleanup schemes

The two schemes were designed in an earlier study [2]. That study used the previous, Julia-based version of the OBE code and treated CU2 at the original field of 250 V/cm. Some of its assumptions differ from the present baseline: other laser powers and beam heights, and 5 % loss per pass in its final comparison. Its numbers are quoted in this chapter because they explain the choices. The results for the present baseline are in Chapters 5 and 6.

3.1 Dark states and how to avoid them

For a fixed laser polarization and frequency, N_g ground sublevels coupled to N_e excited sublevels leave at least $N_g - N_e$ ground-state superpositions that do not couple to the light. Molecules in them are never pumped. In the field, N_e counts only the excited sublevels that lie within reach of the laser spectrum, because the Stark splitting moves the others tens of MHz away. Three tools reduce the problem, and both schemes use all of them:

1. **reach enough excited sublevels**, with the carrier placed between them, with phase-modulation sidebands, or with a second beam on a second excited hyperfine level;
2. **switch the polarization** between X and Z at 500 kHz, so that a superposition dark for one polarization is bright for the other;
3. **use field mixing**, which makes zero-field-forbidden transitions driveable (Sec. 2.4).

3.2 CU1: one laser without phase modulation

At 150 V/cm the five sublevels of B, $J' = 2$, $F_1' = 5/2$, $F' = 2$, $P' = +$ form three groups: $m_F' = 0$ highest, $|m_F'| = 1$ 17.4 MHz below it and $|m_F'| = 2$ a further 24.4 MHz below. The 12 sublevels of $J = 1$ span 2.4 MHz (Fig. 3.1).

Starting idea. One laser with a resonant EOM whose two first-order sidebands reach the $m_F' = 0$ and $|m_F'| = 2$ sublevels: $\Delta_{\mathrm{EOM}} = (17.4 + 24.4)/2 = 20.9$ MHz, with the carrier 3.5 MHz below the $|m_F'| = 1$ pair.

Result. Scans of laser detuning against EOM frequency at 10, 35, 60, 85 and 110 mW (Fig. 3.2) put the minimum of the remaining population at *zero* modulation frequency for every power, at a detuning of about +20 MHz. A single unmodulated frequency near the $|m_F'| = 1$ pair is enough; the outer sublevels need not be addressed directly. The $F = 0$ sublevel, which cannot reach $F' = 2$ at zero field, is pumped through its field-induced $F = 2$

admixture (Table 2.3). All 12 sublevels are emptied within the first $\sim 2\,\mathrm{cm}$ of the interaction region.

Beam height. At 80 mW with a $3\,\mathrm{mm} \times 2.4\,\mathrm{cm}$ beam the cleanup is complete for $|y_0| \lesssim 0.6\,\mathrm{cm}$, but the remaining population rises to about 0.12 at $|y_0| = 1.2\,\mathrm{cm}$. With a 4.8 cm beam it is 0.005 on axis and 0.028 at $|y_0| = 1.2\,\mathrm{cm}$. CU1 uses the 4.8 cm beam.

Opposite parity. With the Λ-doublet partner included, the detuning scan shows a second group of peaks, from the opposite-parity component, between +50 and +80 MHz; the optimum at +20 MHz is unchanged (Chapter 5).

Setup. Frequency-doubled laser → polarization EOM (500 kHz) → multipass (Fig. 3.5a).

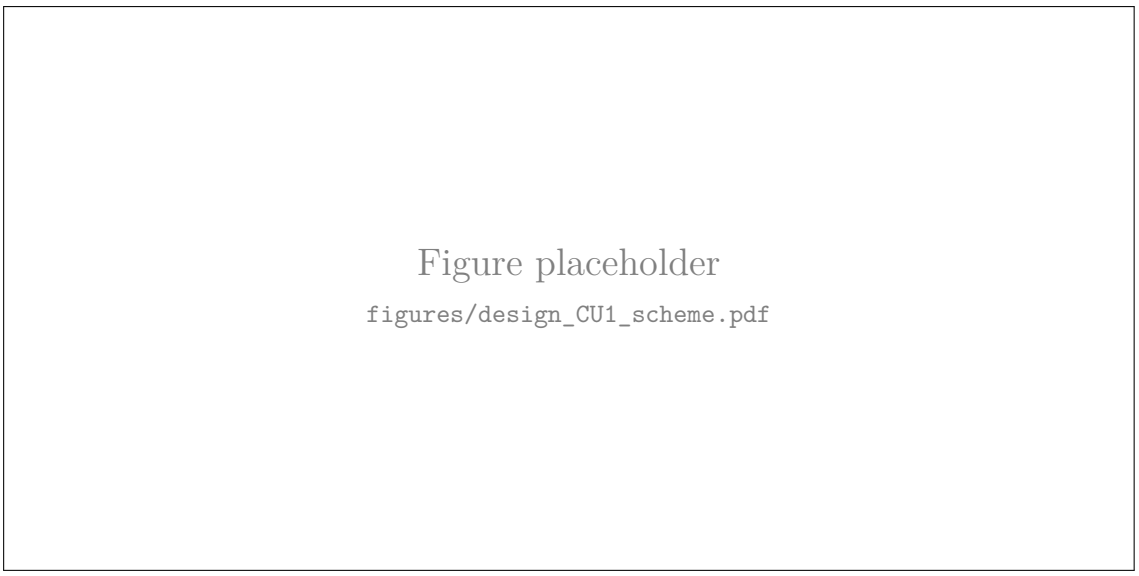


Figure 3.1: CU1 design. (a) Levels of X, $J = 1$ and of B, $J' = 2$, $F_1' = 5/2$, $F' = 2$ at 150 V/cm. (b) Simplified scheme: five excited sublevels split by 17.4 and 24.4 MHz, 12 ground sublevels within 2.4 MHz, and the initial EOM idea. Design-study slides 3–5 [2].

3.3 CU2: from one laser to two hyperfine levels

At 250 V/cm the sublevels of B, $J' = 3$, $F_1' = 7/2$, $F' = 3$, $P' = -$ are split by about 18 MHz per step in $|m_F'|$, and the 20 sublevels of $J = 2$ span 1.4 MHz. The scheme was developed in the steps of Table 3.1.

1. **One laser on $F' = 3$ with a resonant EOM.** The best point (EOM ≈ 17 MHz, $\beta \approx 1.4$–1.6, detuning ≈ 18.5 MHz, 100 mW) cleans only about 98 %. The bottleneck is $|J = 2, F_1 = \frac{3}{2}, F = 1, m_F = 0\rangle$, which keeps about a fifth of its population. An $F = 1$ sublevel reaches $F' = 3$ only through its admixture of $F = 3$, and $m_F = 0$ reaches only $m_F' = \pm 1$, through the EOM sidebands.

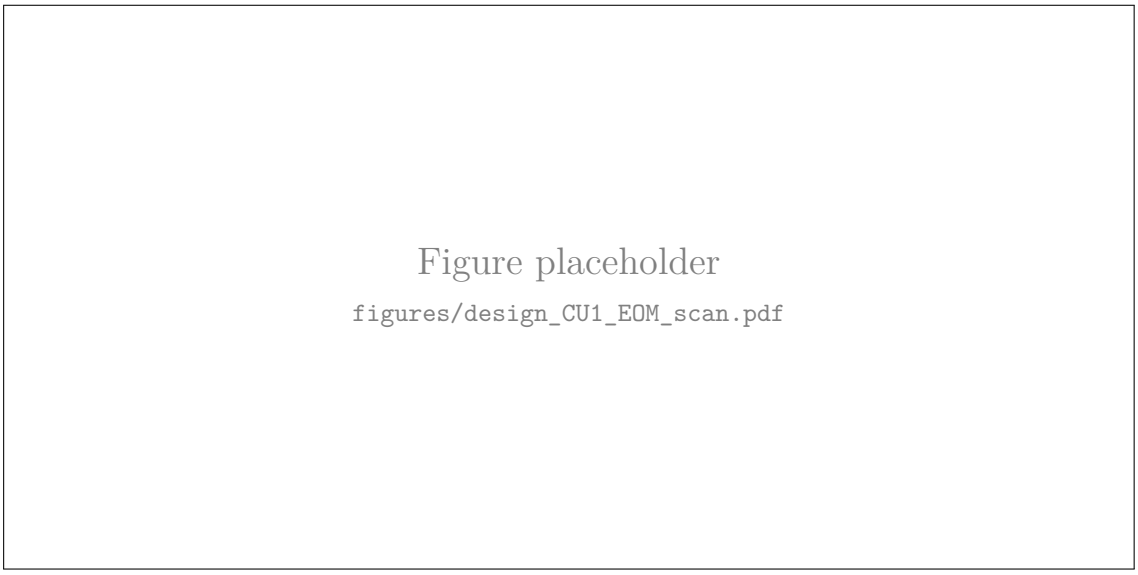


Figure 3.2: CU1 design: remaining $J = 1$ population versus laser detuning and EOM frequency at 10, 35, 60, 85 and 110 mW. The minimum lies at zero modulation and about +20 MHz at every power. Design-study slides 6 and 8 [2].

2. **A weak second tone on $F' = 3$, $m'_F = 0$, made with an AOM.** With 10 mW and Z polarization it helps only at a single, narrow detuning, where the remaining population falls from 1.35 % to 0.76 %. Three ground sublevels couple to a single excited sublevel and quickly form a dark state, and with a fixed Z polarization the polarization switching is not used.

3. **Second beam on $F' = 4$ instead.** The $F' = 4$ level lies about 200 MHz above $F' = 3$. With 70 mW on $F' = 3$ and 10 mW on $F' = 4$, the scan of the second beam shows dips at the $F' = 4$ m'_F components (spaced by 11 and 15 MHz near the top of the manifold), with 0.6 % remaining at the best one, and the $m_F = 0$ bottleneck is removed with the same total power. The cost is a UV AOM at about 200 MHz with some tuning range.

4. **Phase-modulate the second beam as well.** A second EOM on the $F' = 4$ beam (≈ 11.5 MHz, $\beta = 2$) raises the cleanup above 99.9 % (70 + 10 mW). It stays above 99 % at 50 + 7 mW and reaches 99.95 % at 60 + 18 mW: the scheme works better with more power in the second beam.

5. **One EOM for both beams.** With 17.5 MHz, $\beta = 1.5$ on both beams (60 + 18 mW) the cleanup is still about 99.6 %. A single EOM placed before the AOM is much simpler and was adopted.

6. **Final comparison** (single EOM, 5 % loss per pass, 50 + 20 mW): 99.8 % with a 4 cm high beam, 99.95 % with 3 cm.

7. **Present code with the Λ-doublet partners**, 80 + 24 mW, 2.4 cm, 10 % loss: 99.99 % at the chosen point $(\delta_0, \delta_1)/2\pi = (+18.5, -6.5\,\text{MHz})$ (Appendix B). A two-dimensional detuning map at this setting shows a lattice of optima, with the 17.5 MHz EOM period along the $F' = 3$ axis.

Table 3.1: CU2 design steps at 250 V/cm (design study [2], except the last row). Powers are $F' = 3$ + second beam.

step	beams	phase modulation	cleanup
1	$F' = 3$, 100 mW	one EOM, ≈ 17 MHz, $\beta \approx 1.5$	$\approx 98\,\%$
2	$+ F' = 3$, $m'_F = 0$ tone, 10 mW, Z	second tone unmodulated	99.24 % at one narrow detuning
3	$F' = 3 + F' = 4$, 70 + 10 mW	EOM on $F' = 3$ only	$\approx 99.4\,\%$; $m_F = 0$ bottleneck removed
4	70 + 10 mW / 50 + 7 mW / 60 + 18 mW	+ EOM2 ≈ 11.5 MHz, $\beta = 2$ on $F' = 4$	$> 99.9\,\%$ / $> 99\,\%$ / 99.95 %
5	60 + 18 mW	one EOM, 17.5 MHz, $\beta = 1.5$, both beams	$\approx 99.6\,\%$
6	50 + 20 mW, 4 / 3 cm, 5 % loss	as 5	99.8 % / 99.95 %
7	80 + 24 mW, 2.4 cm, 10 % loss	as 5; present code, both parities	99.99 %

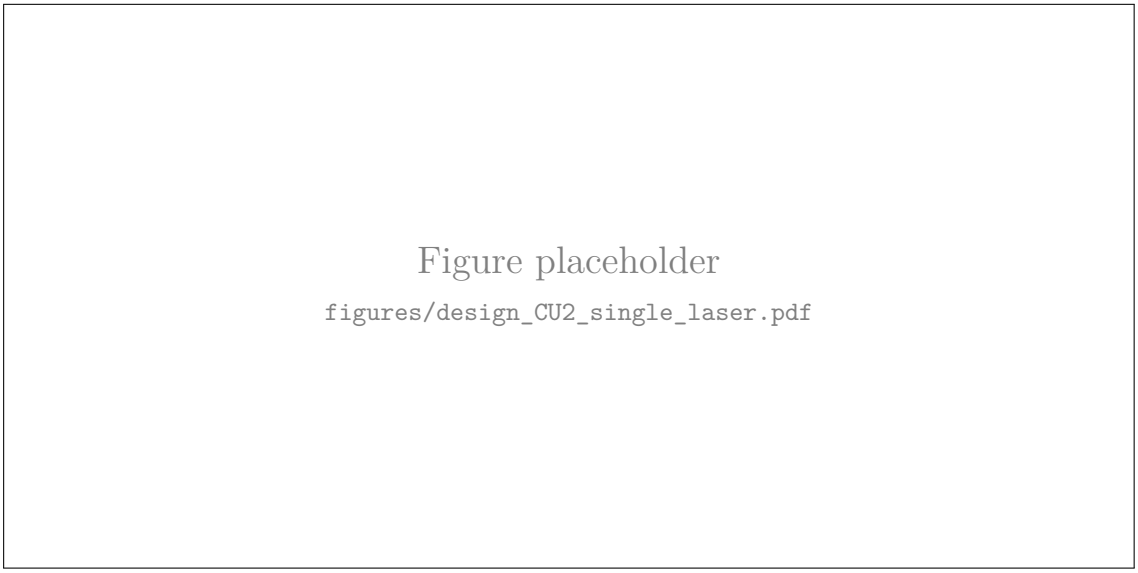


Figure 3.3: CU2 design, step 1 (250 V/cm, 100 mW). (a) Remaining $J = 2$ population versus detuning and EOM frequency. (b) Sublevel populations along the interaction region: $|F_1 = \frac{3}{2}, F = 1, m_F = 0\rangle$ is not cleaned. Design-study slides 17 and 19 [2].

3.4 Optical layouts

The adopted layouts are shown in Fig. 3.5. For CU1 the frequency-doubled laser passes the polarization EOM and enters the multipass. For CU2 the laser first passes the phase EOM (17.5 MHz), so both beams carry the same modulation. A single-pass AOM (around 200 MHz) then produces the two beams from its undiffracted and first-order outputs, so their frequency difference equals the AOM frequency. They are recombined with orthogonal polarizations on a polarizing beam splitter and pass a common polarization EOM (500 kHz) before the multipass. One beam is therefore always Z-polarized while the other is X-polarized.

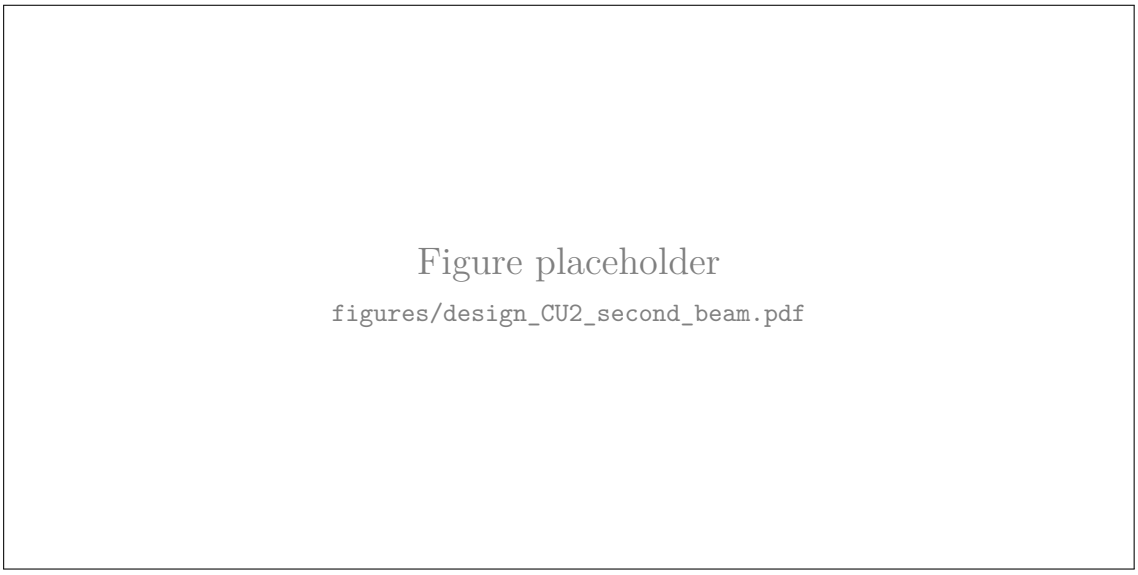


Figure 3.4: CU2 design, steps 3–5. (a) Remaining $J = 2$ population versus the detuning of the $F' = 4$ beam (70 + 10 mW), with the dips assigned to $F' = 4$ sublevels. (b) Detuning of the $F' = 4$ beam versus frequency of its EOM. Design-study slides 23–27 and 31 [2].

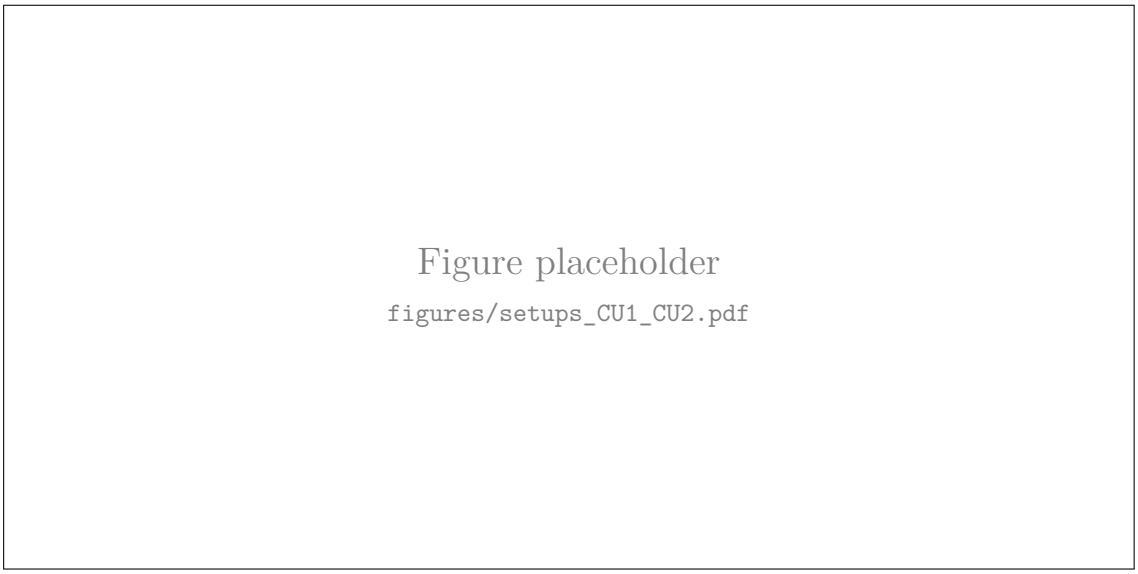


Figure 3.5: Optical layouts. (a) CU1: laser → polarization EOM → multipass. (b) CU2: laser → phase EOM (17.5 MHz) → AOM (≈ 200 MHz, both outputs used) → polarizing beam splitter → polarization EOM (500 kHz) → multipass. Design-study slide 33 [2].

3.5 Operating constraints

3.5.1 AOM coupling of the CU2 beams

Because both CU2 beams come from one laser, their detunings are not independent. With a single-pass AOM, the AOM frequency is the frequency difference of the two beams, Eq. (2.2):

$$f_{\mathrm{AOM}} = \Delta_{\mathrm{main}}(E) + \frac{\delta_1 - \delta_0}{2\pi}, \qquad \frac{\delta_1}{2\pi} = \frac{\delta_0}{2\pi} + f_{\mathrm{AOM}} - \Delta_{\mathrm{main}}(E). \tag{3.1}$$

The laser frequency sets δ_0, and the AOM frequency sets $\delta_1 - \delta_0$. The AOM covers $f_{\mathrm{AOM}} = 200 \pm 15\,\mathrm{MHz}$, so only a band of the (δ_0, δ_1) plane is reachable (Table 3.2). At 200 V/cm it is $-26.83 \le (\delta_1 - \delta_0)/2\pi \le +3.17\,\mathrm{MHz}$.

Table 3.2: Relation between AOM frequency and the detuning difference of the two CU2 beams, Eq. (3.1), for the AOM range 185–215 MHz.

		$(\delta_1 - \delta_0)/2\pi$		f_{AOM} for
E	Δ_{main}	$f_{\mathrm{AOM}} = 185\,\mathrm{MHz}$	$f_{\mathrm{AOM}} = 215\,\mathrm{MHz}$	$(\delta_1 - \delta_0)/2\pi = -25\,\mathrm{MHz}$
200 V/cm	211.83 MHz	−26.83 MHz	+3.17 MHz	186.83 MHz
250 V/cm	219.93 MHz	−34.93 MHz	−4.93 MHz	194.93 MHz

Two consequences follow. First, the former set point $(+18.5, -6.5\,\mathrm{MHz})$ was found at $f_{\mathrm{AOM}} = 194.93\,\mathrm{MHz}$. At 200 V/cm the same AOM frequency gives $(\delta_1 - \delta_0)/2\pi = -16.9\,\mathrm{MHz}$, while keeping the same detunings requires $f_{\mathrm{AOM}} = 186.83\,\mathrm{MHz}$, close to the lower end of the range. Second, every scan for CU2 is naturally a scan of laser frequency and AOM frequency. Scanning δ_1 at fixed δ_0 is a scan of f_{AOM}, and the correlated scans, in which both detunings move together, are scans of the laser frequency at fixed f_{AOM}.

3.5.2 Common frequency shifts

Anything that shifts both beams relative to the molecular lines by the same amount, such as a change of the laser frequency or of the electric field, moves the operating point along the correlated direction. In the original design the field varies by at most about 2 V/cm across the cleanup region, which shifts the lines by about 2 MHz (Fig. 3.6). The width of the optimum along this direction is therefore a tolerance requirement (Chapters 5 and 6). Transverse velocity also shifts the lines, by 3.68 MHz per m/s in the UV, but not as a common shift: in the multipass its sign alternates from one pass to the next (Sec. 4.7), so it cannot be compensated by a change of laser frequency. Its effect is studied for CU1 in Secs. 5.6–5.10 and for CU2 in Sec. 6.7.

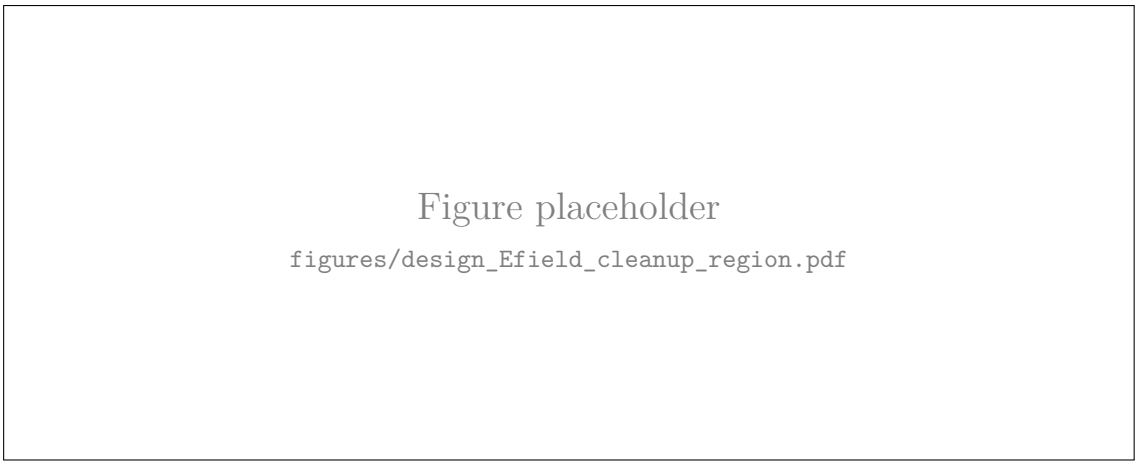


Figure 3.6: Electric field along the beam axis in the cleanup region (original 250 V/cm design); the variation over the laser region is at most about 2 V/cm, i.e. about 2 MHz. Design-study slide 41 [2].

3.5.3 Beam size

The design study scanned the transverse position of the molecule against the beam height for both stages. For CU2 it concluded that the largest available cylindrical lens covers molecules within about $\pm 1.5\,\mathrm{cm}$. In its two-dimensional maps (slides 11 and 42) the on-axis cleanup does not depend on the beam height. This is the signature of an implementation feature shared with the present notebooks (Sec. 5.5): the height is varied at constant peak intensity, not at constant power. The coverage of large beams is therefore overestimated, and the maps should be repeated at constant power. The one-dimensional CU1 comparison of 2.4 and 4.8 cm (Sec. 3.2) does show the expected on-axis loss for the larger beam. The baseline of this report fixes the CU2 beam height at 2.4 cm.

Chapter 4

Simulation

This chapter describes the simulation completely: which states are kept and why, how the Hamiltonian and the decay are built, how the laser light is modelled, what the initial state is, how the master equation is integrated, what is measured, and how scans are run and stored. Section 4.12 collects every assumption with its justification, and Appendix A lists every parameter.

4.1 Overview and workflow

Each cleanup configuration is simulated in five steps (Fig. 4.1). The expensive steps (1–3) are carried out once per configuration; the cheap steps (4–5) are repeated for every point of a scan.

1. **Transition selection.** Choose the transition(s), the polarizations of each beam and the main state pair of each beam (`couplings.generate_transition_selectors`).
2. **OBE system.** Diagonalise the X and B Hamiltonians in the applied fields, identify the eigenstates, select the states coupled by the light, compact the uncoupled rotational levels, build the coupling matrices, the symbolic rotating-frame Hamiltonian and the collapse operators (`lindblad.generate_OBE_system_transitions`).
3. **Runtime parameters and preparation.** Define every time dependence of the drive (multipass beam, phase modulation, polarization switching) and every quantity that will be scanned as a named runtime parameter (`lindblad.LindbladParameters`), bind them to the symbols of the Hamiltonian, and lower the problem to the Rust backend (`lindblad.prepare_lindblad_problem`).
4. **Initial state.** Build the initial density matrix by quantum-number labels.
5. **Integration and analysis.** Integrate the master equation through the interaction region, for one trajectory (`lindblad.solve_lindblad`) or for a grid of parameter values (`lindblad.parameter_scan` in chunks), compute the cleanup efficiency and photon number, and store scans in HDF5 with their full metadata.

4.2 Software

All results use `centrex-tlf` 0.2.5 [3] with its compiled Rust Lindblad backend, in the conda environment `CeNTREX_state_cleanup_new` (Python 3.11). The notebooks place the installed 0.2.5 package ahead of the source checkout on the Python path and check the version

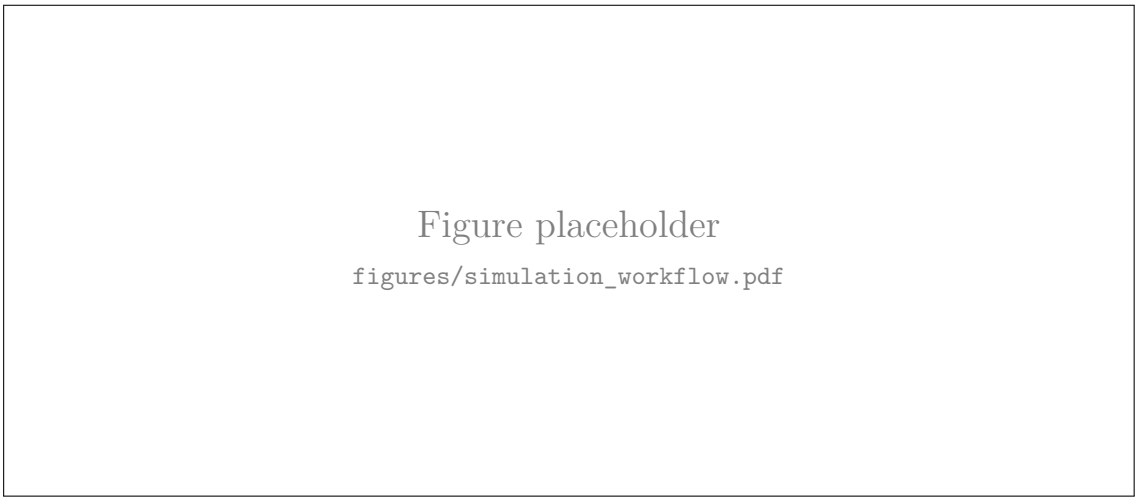


Figure 4.1: Simulation workflow. Once per configuration: transition selection, field-dressed diagonalisation and state identification, state selection and compaction, coupling matrices, rotating-frame Hamiltonian, collapse operators and lowering to the Rust backend. Every scan point: runtime parameters (detunings, AOM frequency, geometry), integration through 5 cm at 184 m/s, cleanup efficiency and photon number, HDF5 storage.

at start-up. They were migrated from the earlier workflow based on the Julia extension `centrex_tlf_julia_extension`, which was used for the design study [2] and is no longer needed.

4.3 Basis, Hamiltonian and state identification

Fields. The electric field is $\mathbf{E} = (0, 0, E)$, with $E = 150\,\mathrm{V/cm}$ for CU1 and $E = 200\,\mathrm{V/cm}$ for CU2. A magnetic field $\mathbf{B} = (0, 0, 1 \times 10^{-3}\,\mathrm{G})$ is added. It has no physical effect on the cleanup but lifts the degeneracy of the $\pm m_F$ pairs. At smaller fields the eigenvectors within a $\pm m_F$ pair are numerically ill-determined: their mixing angle depends on the linear-algebra library at the 10^{-4}–10^{-2} level, which matters because X-polarized light addresses both members of a pair.

Hamiltonians. The X and B effective Hamiltonians of `centrex-tlf` include rotation (for X with the quartic centrifugal-distortion term), the hyperfine and nuclear spin–rotation interactions, Λ-doubling of B, and the Stark and Zeeman interactions. They are built fresh for every configuration. Cached Hamiltonians from earlier package versions are not used, because B_{rot} of X and the centrifugal term have changed by tens of kHz.

Rotational truncation. X is diagonalised over $J = 0 \ldots J_{\max}$ and B over $J' = 1 \ldots J_{\max}$, each in the full hyperfine basis and, for B, with both parities. The default is $J_{\max} = J' + 2$, which gives $J_{\max} = 4$ for CU1 ($J' = 2$) and $J_{\max} = 5$ for CU2 ($J' = 3$), where it is also set explicitly. At fields of a few hundred V/cm this truncation is converged: raising it shifts X energies by $\sim 0.03\,\mathrm{kHz}$ and leaves the main matrix element unchanged to six digits.

State identification. Each field-dressed eigenvector is assigned the bare label with which it has the largest overlap (`states.find_exact_states`). Field mixing makes the labels impure, but the assignment stays correct as long as the best overlap is clearly separated from the second best. For B with both parities retained, one-shot assignment has been verified against adiabatic continuation from zero field up to 200 V/cm; above about 500 V/cm it fails. The 150 V/cm and 200 V/cm builds issue no ambiguity warning. A 250 V/cm build of a single-transition CU2 variant (Appendix B) does warn: two states are assigned with overlap 0.391 against a runner-up of 0.379.

4.4 Reduced model: kept states and compaction

The full diagonalised basis contains hundreds of states, most of which never interact with the light. The OBE system keeps:

- **all sublevels of the addressed ground rotational level** (12 for $J = 1$, 20 for $J = 2$), as individual levels with all coherences;
- **the addressed excited hyperfine level(s)**, all m'_F sublevels, and, if requested, the opposite-parity Λ-doublet partner of each, again with all coherences;
- **every other rotational level of X reachable by decay**, each *compacted* into a single level (`qn_compact=True`).

Compaction is exact for the quantity of interest. The compacted levels are not addressed by any laser, so their internal structure and coherences cannot affect the dynamics; only their total population, fed by decay, matters. All other B levels enter only through the diagonalisation, i.e. through the mixing they produce in the kept states, and are not dynamical levels. Table 4.1 lists the resulting model sizes. All CU2 results use the 56-level model with both Λ-doublet components retained.

Table 4.1: Sizes of the OBE models. "Compacted" are the other rotational levels of X, one level each.

model	levels	X levels	B levels
CU1, addressed parity only	21	12 ($J = 1$) + 4 compacted ($J = 0, 2, 3, 4$)	5 ($J' = 2$, $F' = 2$, $P' = +$)
CU1, both parities	26	12 + 4 compacted	10 ($F' = 2$, both P')
CU2, two transitions	56	20 ($J = 2$) + 4 compacted ($J = 1, 3, 4, 5$)	32 ($F' = 3$: 2×7; $F' = 4$: 2×9)
CU2 variant, single transition	38	20 + 4 compacted	14 ($F' = 3$: 2×7)

4.5 Optical couplings and the rotating frame

Coupling matrices. For each beam k and each polarization $p \in \{X, Z\}$, with $\hat{\boldsymbol{\epsilon}}_X = \hat{\mathbf{x}} \perp \mathbf{E}$ and $\hat{\boldsymbol{\epsilon}}_Z = \hat{\mathbf{z}} \parallel \mathbf{E}$, the coupling matrix

$$M^{(p)}_{k,eg} = \frac{\langle e| \, \mathbf{d} \cdot \hat{\boldsymbol{\epsilon}}_p \, |g\rangle}{D_{XB}} \tag{4.1}$$

is evaluated between the field-dressed states, for the ground states of the addressed level and the excited states of beam k. Whether a pair is coupled is decided by the magnitude of this mixed-state matrix element, not by the zero-field selection rules: in the field, nominally forbidden pairs can be genuinely driven (Sec. 2.4). Z light drives $\Delta m_F = 0$, X light a superposition of $\Delta m_F = \pm 1$.

Normalisation. The Rabi rate Ω_k of beam k refers to its main pair: each matrix is divided by the main-pair element $m_k = M^{(X)}_{k,e_0 g_0}$, so that the main pair is driven at exactly Ω_k and every other pair at $\Omega_k M_{eg}/m_k$. The conversion from laser intensity to Ω_k uses the same m_k (Sec. 4.7). The result is therefore independent of which pair is chosen as main, provided the conversion and the Hamiltonian use the same, freshly computed m_k, and provided m_k is not a weak, mixing-only element (`centrex-tlf` warns below 1 % of the strongest element).

Hamiltonian. In the rotating-wave approximation and with $\hbar = 1$ (all energies are angular frequencies in rad/s),

$$H(t) = H_0 - \sum_k \delta_k \, \Pi_k + \frac{1}{2} \sum_k \Big[\Omega_k(t) \sum_{p \in \{X,Z\}} g_{p,k}(t) \, \frac{M^{(p)}_k}{m_k} + \text{h.c.} \Big], \tag{4.2}$$

where H_0 holds the field-dressed level energies in the rotating frame, Π_k projects onto the excited states in the frame of beam k, $g_{p,k}(t)$ is the polarization gate (Sec. 4.7), and $\Omega_k(t)$ is complex when phase modulation is used. The detuning of beam k is measured from its main resonance:

$$\omega_k = \left(E_{e_0} - E_{g_0}\right)_k + \delta_k, \tag{4.3}$$

so $\delta_k > 0$ is a blue detuning. Since the main resonance itself moves with E, a fixed δ_k is a different laser frequency at a different field.

Two beams in one frame. In the CU2 model the 32 excited levels are put in the rotating frame of beam 0. Beam 1 then carries an explicit phase factor: its couplings contain $\exp[\mathrm{i}(\omega_1 - \omega_0)t] = \exp(2\pi \mathrm{i} \, f_{\mathrm{AOM}} t)$, at 185–215 MHz. This is an exact change of representation, and it is why δ_1 enters the Hamiltonian only through this phase. The integrator, however, must resolve a period of about 5 ns, which is likely an important contributor to the much larger cost of a CU2 trajectory (Sec. 4.9).

Cross-coupling. Each beam couples only to its own excited level; the off-resonant action of beam 0 on $F' = 4$ and of beam 1 on $F' = 3$ is neglected. In the frame of beam 0, the $F' = 3$ components lie between 0 and 87 MHz and the $F' = 4$ components between 180 and 267 MHz. With phase-modulation sidebands up to third order (0.4 % of the power each), beam 0 reaches at most about 71 MHz for $\delta_0/2\pi \leq 18.5$ MHz. Beam 1 lies at $\delta_0/2\pi + f_{\mathrm{AOM}}$, so its third-order sidebands come within about $f_{\mathrm{AOM}} - 52.5\,\mathrm{MHz} - 87\,\mathrm{MHz} + \delta_0/2\pi$ of the highest $F' = 3$ component. Over the whole two-dimensional scan ($\delta_0/2\pi \geq 5$ MHz, $f_{\mathrm{AOM}} \geq 185$ MHz) all neglected couplings are detuned by more than 50 MHz, and by 63 MHz at the optimum. That is more than 31 Γ and more than 65 times the largest Rabi rate, which suppresses their excitation probability to below 10^{-4}.

4.6 Spontaneous decay

Every excited state decays with the population decay rate $\Gamma = 1/\tau = 1.0101 \times 10^7\,\mathrm{s}^{-1}$ ($\Gamma/2\pi = 1.608$ MHz), from the measured lifetime $\tau = (99 \pm 9)$ ns. The branching ratios $b_{e\to g}$ are computed from the dipole matrix elements between the field-dressed states, to every X state of the diagonalised basis, and normalised to one. Decays into a compacted rotational level are summed into that level. One collapse operator is used per excited–ground pair,

$$C_{eg} = \sqrt{\Gamma\, b_{e\to g}}\; |g\rangle\langle e|, \tag{4.4}$$

and the density matrix obeys the Lindblad master equation [4]

$$\dot{\rho} = -\mathrm{i}\,[H(t), \rho] + \sum_{e,g} \Big(C_{eg}\,\rho\,C_{eg}^{\dagger} - \tfrac{1}{2}\{C_{eg}^{\dagger} C_{eg}, \rho\} \Big). \tag{4.5}$$

Because the basis contains every rotational level that the addressed excited states can decay to, the system is closed and $\mathrm{Tr}\,\rho = 1$ is conserved. Two simplifications follow from this form:

- **No decay-induced ground-state coherences.** With one operator per pair, spontaneous emission transfers populations but not coherences between ground sublevels; that would require one operator per emitted polarization component. The cleanup efficiency depends on total populations, so the effect is expected to be small, but it is not zero where dark superpositions matter.
- **No vibrational loss.** Decay to $X(v \geq 1)$ is not modelled (`decay_channels=None`). A molecule that decays to $v \geq 1$ is removed from the addressed level just as one that decays to another J, whereas the model returns part of that decay to the addressed level. The neglect can therefore only underestimate the cleanup.

4.7 Laser field model

Coordinates. The molecule moves along the beam coordinate s (called z in the notebooks, e.g. `vz`, `sigma_z`) with $s = vt$; y is the transverse coordinate along which the laser beam is

elongated (its height); the molecule sits at a fixed transverse offset y_0 ($y_0 = 0$ unless scanned). The light crosses the molecular beam transversely, along x. Its direction enters only through the Doppler shift of a molecule with transverse velocity v_x (paragraph *Doppler shift* below).

Multipass geometry. The light crosses the molecular beam 13 times in a two-prism multipass. The pass centres are at $s_n = 3.5\,\mathrm{mm} \times n$, $n = 1 \ldots 13$ (3.5 mm to 45.5 mm, spaced by 3.5 mm, i.e. 19 µs at 184 m/s), and $y_n = 0$. With a loss $L = 10\,\%$ per pass, the pass that is crossed p-th in optical order carries the power fraction $(1 - L)^{p-1}$. The spatial order of the optical passes in a two-prism multipass is `utils.multipass.multipass_prism_order(13)` $= (1, 12, 3, 10, 5, 8, 7, 6, 9, 4, 11, 2, 13)$, so along the molecular beam the relative powers are

$$a_{1\ldots7} = (1.000,\ 0.314,\ 0.810,\ 0.387,\ 0.656,\ 0.478,\ 0.531),$$
$$a_{8\ldots13} = (0.590,\ 0.430,\ 0.729,\ 0.349,\ 0.900,\ 0.282),$$

which sum to 7.46. Each pass is an elliptical Gaussian with $\sigma_s = 0.75\,\mathrm{mm}$ along the beam (full 4σ width 3 mm) and $\sigma_y = 12\,\mathrm{mm}$ (CU1) or 6 mm (CU2) transversely (beam heights $4\sigma_y$ of 4.8 cm and 2.4 cm). The intensity profile along the beam is shown in Fig. 4.2.

Intensity and Rabi rate. The passes add incoherently in intensity,

$$I_k(s, y) = P_k \sum_{n=1}^{13} \frac{a_n}{2\pi\sigma_s\sigma_y} \exp\Big[-\frac{(s - s_n)^2}{2\sigma_s^2} - \frac{(y - y_n)^2}{2\sigma_y^2}\Big], \tag{4.6}$$

with P_k the optical power of beam k entering the multipass, so interference between passes is neglected. The Rabi rate of the main pair is

$$|\Omega_k(t)| = \frac{|m_k|\, D_{XB}}{\hbar} \sqrt{\frac{2I_k(vt, y_0)}{c\,\varepsilon_0}}, \qquad D_{XB} = 2.667\,550\,6 \times 10^{-30}\,\mathrm{C\,m}. \tag{4.7}$$

In the code the profile is evaluated for $P = 1\,\mathrm{W}$ (runtime parameter `intensities_per_watt`) and multiplied by $\sqrt{P_k}$, which is exact because $\Omega \propto \sqrt{I}$. The value of D_{XB} is passed explicitly in the notebooks; it is 0.12 % below the `centrex-tlf` default `ED_XtB` $= 0.315\,ea_0$. Table 4.2 lists the resulting peak values.

Table 4.2: Peak intensity (strongest pass, $a_n = 1$, on axis) and main-pair Rabi rate of each beam, before phase modulation, and the on-resonance saturation parameter $s_0 = 2|\Omega|^2/\Gamma^2$.

beam	P	I_{peak}	$\lvert\Omega\rvert/2\pi$	$\lvert\Omega\rvert/\Gamma$	s_0
CU1, R(1) $F' = 2$	80 mW	0.141 W/cm^2	0.417 MHz	0.26	0.13
CU2, R(2) $F' = 3$	80 mW	0.283 W/cm^2	0.557 MHz	0.35	0.24
CU2, R(2) $F' = 4$	24 mW	0.085 W/cm^2	0.736 MHz	0.46	0.42

Transit time. At 184 m/s a molecule needs 9.6 µs to cross the intensity FWHM of one pass and 271.7 µs for the whole 5 cm.

Polarization switching. For any fixed polarization the number of addressed ground sublevels exceeds the number of excited sublevels they couple to (12 against 5 for CU1; 20 against 7 or 9 for each CU2 beam), so at least the difference in ground-state superpositions is dark (Sec. 3.1). The polarization is therefore switched between Z and X with a square wave,

$$g_Z(t) = \tfrac{1}{2}\big[1 + \operatorname{sgn}\sin(\omega_\mathrm{p} t + \phi_\mathrm{p})\big], \qquad g_X(t) = 1 - g_Z(t), \tag{4.8}$$

at $\omega_\mathrm{p}/2\pi = 0.5\,\mathrm{MHz}$ (1 µs in each polarization, about five switching periods per pass) and $\phi_\mathrm{p} = 0$. Switching is instantaneous, and the power is the same in both polarizations. In CU2 the two beams switch in antiphase: beam 0 uses g_Z for Z, beam 1 uses g_X for Z. This reproduces the optical layout, in which the two beams enter the common polarization EOM with orthogonal polarizations (Sec. 3.4).

Phase modulation. For CU2 the components of each beam spread over about 90 MHz (Table 2.2). The laser is therefore phase-modulated by one EOM before the AOM,

$$\Omega_k(t) \;\to\; \Omega_k(t)\, e^{\mathrm{i}\beta \sin\omega_\mathrm{m} t}, \qquad \omega_\mathrm{m}/2\pi = 17.5\,\mathrm{MHz}, \quad \beta = 1.5, \tag{4.9}$$

with the same modulation, including its phase, on both beams. The power is distributed over sidebands at $n \times 17.5\,\mathrm{MHz}$ with fractions $J_n(\beta)^2$: 0.262 in the carrier, 0.311 in each first-order sideband, 0.054 in each second-order and 0.004 in each third-order sideband. CU1 uses no phase modulation in the baseline; a 1.6 MHz EOM for CU1 is studied as an option in Sec. 5.8, with the same form, $\Omega(t) \to \Omega(t)\, e^{\mathrm{i}\beta \sin(2\pi \times 1.6\,\mathrm{MHz}\, t)}$ (runtime parameters `eom_beta`, default 0, and `eom_frequency`).

Doppler shift. A molecule with velocity v_x along the laser beams sees the light shifted by $k_L v_x$, with $k_L v_x/2\pi = 3.68\,\mathrm{MHz}$ per m/s at 271.7 nm (function `velocity_to_detuning` of `utils.detuning`). In the two-prism multipass the light reverses direction after every crossing, so consecutive optical passes counter-propagate. Because the optical order of the spatially ordered passes, $(1, 12, 3, 10, \ldots)$, alternates between odd and even, the direction of the light also alternates from one spatial pass to the next. The Doppler shift therefore changes sign from pass to pass,

$$\delta_k \;\to\; \delta_k + k_L v_x\, \sigma(vt), \qquad \sigma(s) = (-1)^{\lfloor (s - s_1 + \Delta s/2)/\Delta s \rfloor}, \tag{4.10}$$

with $s_1 = 3.5\,\mathrm{mm}$ the first pass and $\Delta s = 3.5\,\mathrm{mm}$ the pass spacing; σ switches midway between two passes, where the intensity is negligible (`lindblad.alternating_sign`). A molecule thus sees the laser at $\delta + k_L v_x$ in half of the passes and at $\delta - k_L v_x$ in the other half. No laser frequency compensates both, which is why the transverse velocity matters much more than a common frequency shift of the same size. The transverse velocity is the runtime parameter `vx` (built with `doppler=True`). It is used for CU1 (Secs. 5.6–5.10) and for CU2 (Sec. 6.7); in CU2 the two beams share the multipass, so the shift is the same for both beams.

AOM. The frequency difference of the two CU2 beams is the AOM frequency, Eq. (3.1). In the model it enters through $\delta_1 = \delta_0 + 2\pi[f_{\mathrm{AOM}} - \Delta_{\mathrm{main}}(E)]$. The two beam powers are set independently, as the powers that enter the multipass.

Drive of each configuration.

- **CU1:** $\Omega(t) = \sqrt{P}\,\Omega_{1\mathrm{W}}(vt, y_0)$, one beam, no phase modulation, $P = 80\,\mathrm{mW}$ (the notebook default for single trajectories is 40 mW).
- **CU2:** $\Omega_k(t) = \sqrt{P_k}\,e^{\mathrm{i}\beta\sin\omega_{\mathrm{m}}t}\,\Omega_{1\mathrm{W},k}(vt, y_0)$ for $k = 0, 1$, with $P_0 = 80\,\mathrm{mW}$ on $F' = 3$ and $P_1 = 24\,\mathrm{mW}$ on $F' = 4$, both through the same multipass.

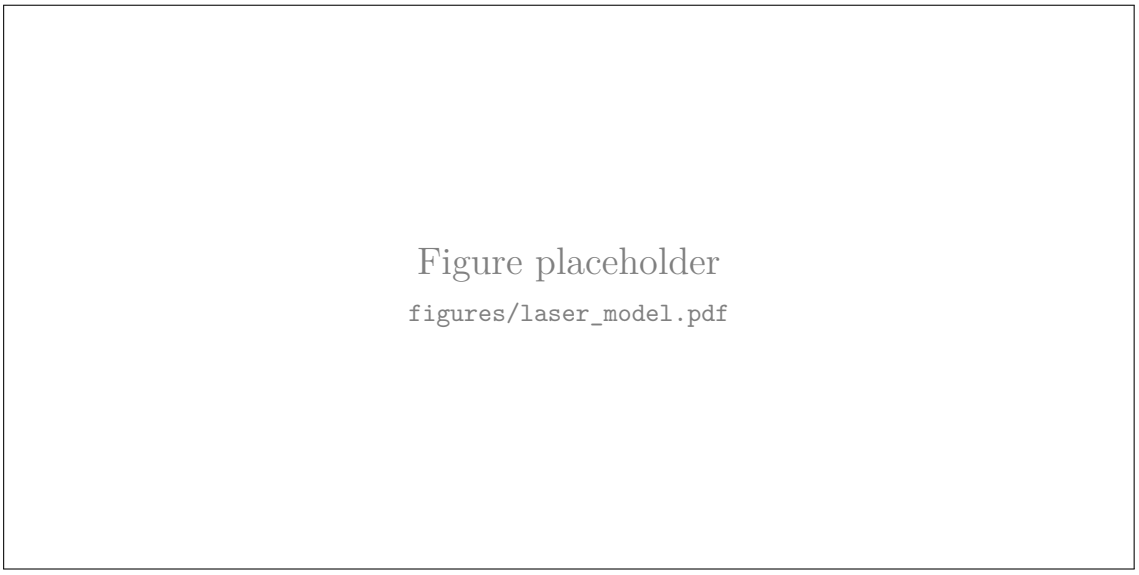


Figure 4.2: Laser model along the trajectory. (a) Intensity of the 13 passes along the beam coordinate for 1 W, with the relative pass powers a_n and the optical pass order. (b) Main-pair Rabi rate seen by the molecule versus time for the CU1 and CU2 beams. (c) Zoom on one pass: polarization gate $g_Z(t)$ at 0.5 MHz and, for CU2, the antiphase gate of the second beam.

4.8 Molecular trajectory and initial state

Trajectory. One molecule moves on the beam axis ($y_0 = 0$) at the mean forward velocity $v = 184\,\mathrm{m/s}$, from $s = 0$ to $s = 5\,\mathrm{cm}$, i.e. for $t \in [0, T]$ with $T = 271.7\,\mu\mathrm{s}$. The intensity is negligible at both ends ($e^{-10.9}$ of a pass peak at $s = 0$, six σ_s from the last pass at $s = 5\,\mathrm{cm}$). The transverse coordinates are constant along the trajectory: at $|v_x| \le 1\,\mathrm{m/s}$ the molecule moves by less than 0.3 mm during T, small against the beam height. The transverse velocity v_x and offset y_0 are scanned, each (v_x, y_0) being one trajectory, and the results are averaged afterwards over the transverse distribution within the detection acceptance (Sec. 5.10). Results labelled *on axis* are for $v_x = 0$, $y_0 = 0$. No average over the forward-velocity spread is taken.

Initial state. The molecules enter with equal, incoherent population in all sublevels of the addressed rotational level,

$$\rho(0) = \frac{1}{N_J} \sum_{i \in J} |i\rangle\langle i|, \qquad N_1 = 12,\ N_2 = 20, \tag{4.11}$$

because the hyperfine splittings are far smaller than the thermal energy of the beam and no earlier stage is assumed to have polarised the sublevels. For CU2 this is a simplification: the $J = 2$ population left after SPB need not be uniform over the sublevels. States are selected by quantum-number label (`QuantumSelector`), never by energy order, because level orderings change with the field.

For CU1 a second initial state is used, the *SPB-related* levels: the eight $J = 1$ eigenstates whose $|m_J| = 1$ content in the uncoupled basis exceeds 0.5, with equal weights. These are the levels into which SPB transfers the molecules, so $J = 1$ population left there before SPB cannot be told apart from the transferred molecules. Because the master equation is linear in $\rho(0)$, one trajectory per starting sublevel gives the result for any incoherent mixture; such per-level results are used in Sec. 5.9.

4.9 Numerical integration

Equation (4.5) is integrated with the adaptive Dormand–Prince 5(4) method (`dopri5`) [5] in the Rust backend, using the `expanded_sparse` execution mode and the `decomposed` Hamiltonian representation, in which $H(t)$ is stored as a constant part plus constant matrices multiplied by scalar time functions. Table 4.3 lists the settings. Single trajectories save the populations every 1 µs (273 time points); scans return only the final populations.

Table 4.3: Integrator settings and cost. Wall-clock times on the analysis laptop, with trajectories of one chunk run in parallel.

run	initial dt	rtol	atol	max. steps	cost
CU1, single and scans	2 ns	10^{-7}	10^{-9}	5×10^5	build 2.6 s; 121-point scan in 64 s (chunks of 32)
CU2, single trajectory	2 ns	10^{-7}	10^{-9}	10^6	build 29 s, preparation 6 s
CU2, scans	2 ns	10^{-6}	10^{-8}	6×10^5	200 V/cm: 42.6 s per trajectory with 8 threads (Table 4.4)
CU2, confirmation	2 ns	10^{-7}	10^{-9}	10^6	95 s per trajectory (3 in parallel)

In the scans a CU2 trajectory took about 80 times longer than a CU1 trajectory. The model has 56 instead of 21 levels, the EOM adds a 17.5 MHz time dependence, and the second beam adds the 185–215 MHz phase factor of Sec. 4.5. Within a scan, trajectories run in parallel only inside one chunk of `parameter_scan`. A chunk size of one serialises the scan, so the chunk size should be well above the number of threads. The number of threads matters as well (Table 4.4). On the analysis machine (Intel i7-12700: 8 performance cores with two threads each and 4 efficiency cores, 20 threads in total) the throughput is highest with 8 threads, one per performance core. Using all 20 threads is 33 % slower, because trajectories

that share a core or run on an efficiency core slow each other down. All production CU2 scans use 8 threads and chunks of 40.

Table 4.4: CU2 scan throughput at 200 V/cm versus the number of Rayon threads (chunks of 40 random grid points, rtol 10^{-6}). The first row was measured with a chunk of 20.

threads	chunk	wall time per trajectory
20	20	59.6 s
20	40	56.7 s
12	40	45.1 s
8	40	42.6 s

4.10 Observables

Cleanup efficiency. The figure of merit is the fraction of molecules removed from the addressed level,

$$\eta_{\mathrm{c}} = 1 - \sum_{i \in J} \rho_{ii}(T), \tag{4.12}$$

with the sum over all sublevels of the addressed ground level. The HDF5 files store the remaining population $1 - \eta_{\mathrm{c}}$.

Photons per molecule. The number of scattered photons is

$$N_\gamma = \Gamma \int_0^T \sum_{e \in B} \rho_{ee}(t)\, \mathrm{d}t, \tag{4.13}$$

evaluated with the trapezoidal rule on the 1 µs output grid. The excited population varies on the 1 µs polarization period and, for CU2, on the 57 ns EOM period, so N_γ obtained this way is approximate. The solver's built-in running integral (`output="photon_integral"`) is the accurate alternative.

Destination of the removed molecules. The final populations of the compacted levels give the distribution of the removed molecules over the other rotational levels.

4.11 Parameter scans and data storage

Detunings, the AOM frequency (through δ_1), powers, the transverse position y_0 and velocity v_x, the beam widths and the polarization and modulation settings (including the optional CU1 EOM) are runtime parameters. Scanning them requires no rebuild of the OBE system. The electric field, the transitions and the choice of retained levels change the coupling matrices and require a new build. Cartesian scans are run with `scripts/scan_progress.py`, which splits the grid into chunks of `lindblad.parameter_scan` and reproduces the C-order layout of `lindblad.grid_scan`. Scans along a line in the (δ_0, δ_1) plane, such as the laser-frequency scans at fixed AOM frequency, are run directly as parameter batches.

Every scan is stored with `scripts/scan_hdf.py` (schema `centrex-cleanup-scan-v1`) as a group `runs/<name>` containing the result array, one dataset per axis with its unit, and JSON attributes for the fixed physical parameters, the model (notebook, package version, transitions, number of levels, initial manifold) and the solver settings. Frequencies are stored in MHz, the solver receives rad/s. Existing runs are not overwritten unless this is requested explicitly. The runs used in this report are listed in Appendix C.

4.12 Summary of assumptions

Table 4.5: Assumptions of the model, their justification and their expected effect on the cleanup efficiency η_c.

assumption	justification	expected effect
rotating-wave approximation, one rotating frame per beam	optical frequency $\sim 1 \times 10^{15}$ Hz $\gg$ every other rate	none
rotational truncation $J_{max} = J' + 2$	converged at few-hundred V/cm (energies to 0.03 kHz, matrix element to 10^{-6})	none
one-shot state identification	validated up to 200 V/cm for B with both parities; no warning at 150 and 200 V/cm	none at the fields used
uncoupled X levels compacted	no light acts on them	none on η_c
other B levels only through mixing	not resonant with any beam	negligible
Λ-doublet partner retained (CU2)	strong Stark mixing at 200 V/cm	for CU1 it changes η_c by $< 10^{-5}$
no cross-coupling between the two CU2 beams	detuned by > 50 MHz at every scanned point	negligible
one collapse operator per pair	standard form of `centrex-tlf`	small; no decay-induced ground-state coherences
no decay to $v \geq 1$	not modelled	conservative: can only lower η_c
passes add incoherently, Gaussian profiles	separated passes	interference not modelled
10 % loss per pass	baseline of this report	sets the pass powers a_n
instantaneous, power-neutral polarization switching	square-wave gate	ideal switching
identical phase modulation on both CU2 beams	one EOM before the AOM	none
single forward velocity $v = 184$ m/s	mean beam conditions	no average over the forward velocity
(v_x, y_0) grid averaged with the beam-simulation distribution	lens at 25 kV; Gaussian fits to the simulated beam	averages depend on the distribution (Secs. 5.7, 5.10)
detection acceptance applied at the cleanup plane	$\lvert v_x \rvert \leq 1$ m/s, $\lvert y_0 \rvert \leq 8.5$ mm; drift $v_y L/v$ to the detector neglected	the detected population differs somewhat from the truncated Gaussian
Doppler sign alternating from pass to pass	counter-propagating consecutive passes of the two-prism multipass	cannot be compensated by the laser frequency
CU2 optima from the on-axis optimisation	R1, R2 found for 2.4 cm, $v_x = 0$	not re-optimised for 3 cm and the transverse distribution
uniform, incoherent initial population	hyperfine splitting $\ll kT$	none if the incoming sublevels are unpolarised

Chapter 5

CU1 results

Configuration: R(1) $F_1' = 5/2$, $F' = 2$ at 150 V/cm, one beam without phase modulation, beam height 4.8 cm, X/Z switching at 0.5 MHz (Appendix C).

5.1 Single trajectory

At the notebook default (40 mW, $\delta = +20.3$ MHz, 21-level model) the cleanup efficiency is $\eta_c = 98.92\,\%$ with $N_\gamma = 1.36$ photons per molecule. The removed molecules end in $J = 3$ (49.2 %), $J = 2$ (47.1 %), $J = 4$ (2.65 %) and $J = 0$ (0.008 %); 1.08 % remains in $J = 1$. At the optimum of Sec. 5.2 (80 mW, +20 MHz) the efficiency is 99.93 % with $N_\gamma \approx 1.2$. The trace stays one to 4×10^{-15}.

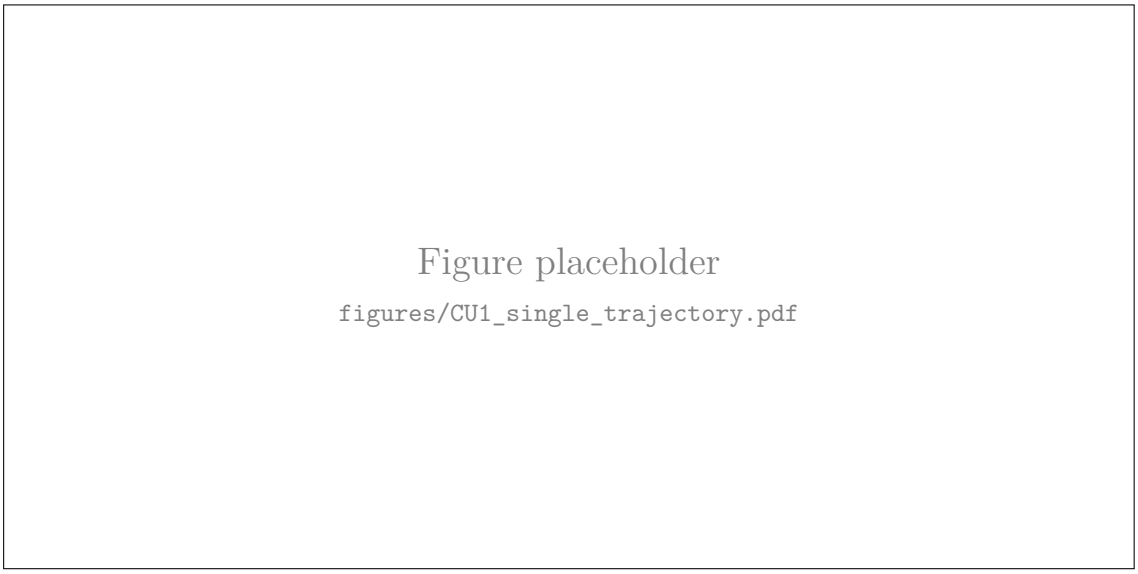


Figure 5.1: CU1, single trajectory at 80 mW, +20 MHz. (a) Populations of the 12 $J = 1$ sublevels versus distance. (b) Total $J = 1$ population, excited-state population and populations of the compacted levels $J = 0, 2, 3, 4$.

5.2 Detuning dependence

Figure 5.2 and Table C.2 summarise the detuning scans at 80 mW. On a 1 MHz grid from −20 to +100 MHz the best point is +20.0 MHz with $\eta_c = 99.9275\,\%$. A 0.1 MHz scan from 19 to 22 MHz locates the optimum at +20.2 MHz with $\eta_c = 99.963\,\%$; the two scans agree exactly at their common points. The optimum coincides with the strength-weighted centre of the

transition components, +19.5 MHz (Table 2.2), as anticipated by the design study (Sec. 3.2). It is narrow (Table 5.1): the cleanup exceeds 99.9 % only between +20.0 and +20.5 MHz, and 99 % between +19.4 and +21.4 MHz. The laser frequency, and the line positions through the electric field, must therefore be held to within about ±0.3 MHz of the optimum to stay above 99.9 %. These are on-axis values for $v_x = 0$; with the transverse velocity distribution the best detuning moves to +19.0 MHz (Sec. 5.7).

Table 5.1: CU1: detuning window within which the cleanup exceeds a given level (80 mW, 0.1 MHz scan, 26-level model).

η_c above	detuning window	width
99.9 %	+20.0 to +20.5 MHz	0.6 MHz
99.5 %	+19.6 to +21.1 MHz	1.6 MHz
99.0 %	+19.4 to +21.4 MHz	2.1 MHz

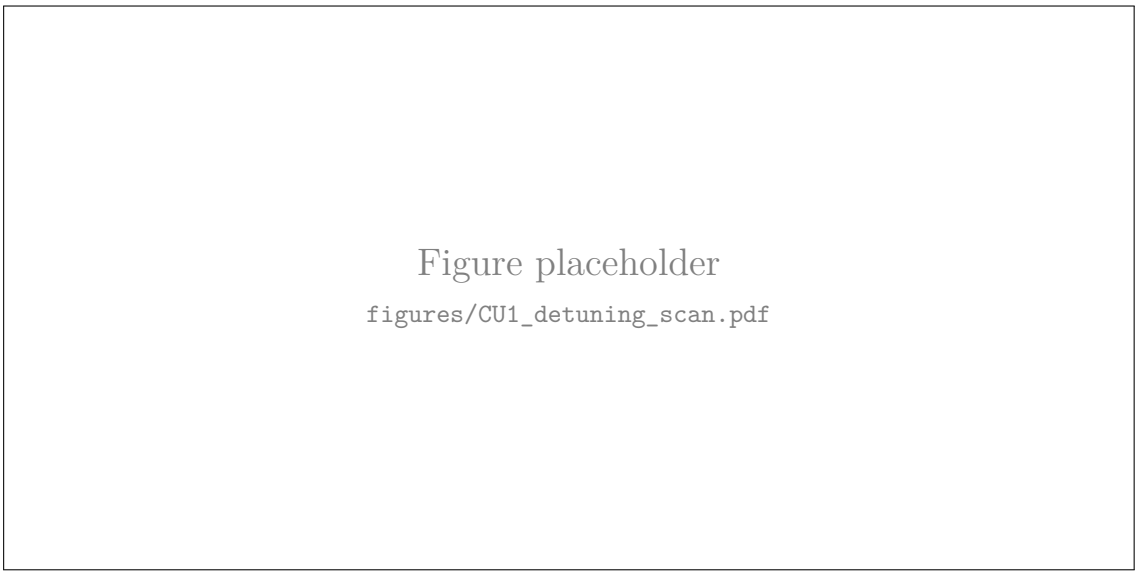


Figure 5.2: CU1 cleanup efficiency versus detuning at 80 mW. (a) −20 to +100 MHz, 1 MHz steps, with and without the Λ-doublet partner (runs `detuning_board_-20to_100MHz` and `detuning_withoutP-_-20to_100MHz`); the partner adds the peaks between +50 and +80 MHz. (b) 19 to 22 MHz, 0.1 MHz steps (run `detuning_fine`), with the windows of Table 5.1.

5.3 Power dependence

Table 5.2 gives the cleanup versus power from a power × detuning scan (10–110 mW, −16 to +30 MHz in 1.02 MHz steps). The best grid point is +20.8 MHz at every power, so the optimum detuning does not depend on power; with the grid spacing this is consistent with the fine-scan optimum at +20.2 MHz. Because the grid misses the exact optimum, the values in the table are lower bounds. Above 50 mW the remaining population falls by a factor of about three per additional 20 mW (1.02, 0.35, 0.12 and 0.045 % at 50, 70, 90 and 110 mW).

Table 5.2: CU1: cleanup efficiency versus laser power at the best detuning of the grid, +20.8 MHz (run `power_detuning_no_phase_eom`, 26-level model).

10 mW	30 mW	50 mW	70 mW	90 mW	110 mW
79.57 %	96.42 %	98.98 %	99.65 %	99.88 %	99.955 %

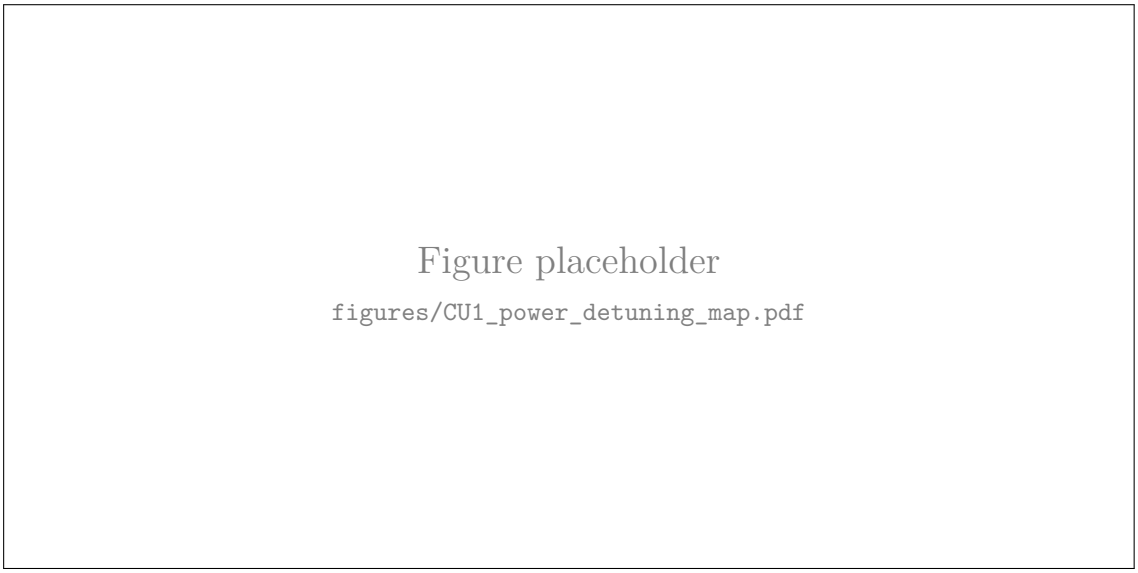


Figure 5.3: CU1 cleanup efficiency (or remaining population on a logarithmic scale) versus laser power and detuning (run `power_detuning_no_phase_eom`).

5.4 Excited-state Λ-doublet partner

Retaining the opposite-parity partner of $F' = 2$ adds 5 levels and 30 coupled pairs, located up to 80 MHz from the main resonance, and changes the branching into the compacted levels. The cleanup at +20 MHz changes from 99.9275 % (21 levels) to 99.9269 % (26 levels), i.e. by 6×10^{-6}. The 21-level model is sufficient for CU1 at 150 V/cm.

5.5 Beam geometry

The transverse position y_0 was scanned from −12 to +12 mm for beam heights $4\sigma_y$ of 10–50 mm, in an earlier configuration (1.0 MHz switching, +20.6 MHz, 80 mW). Near the axis the cleanup is between 99.874 and 99.877 % for every height. This is a consequence of the implementation and not a physical result: the normalisation `intensities_per_watt` $= a_n/(2\pi\sigma_s\sigma_y)$ is computed once with the default σ_y and kept fixed, while the runtime parameter σ_y changes only the width of the Gaussian. The scan therefore varies the beam height at *constant peak intensity*, not at constant power. A scan at constant power requires the normalisation to scale as $1/\sigma_y$, for example by multiplying the drive by $\sqrt{\sigma_y^{\mathrm{ref}}/\sigma_y}$.

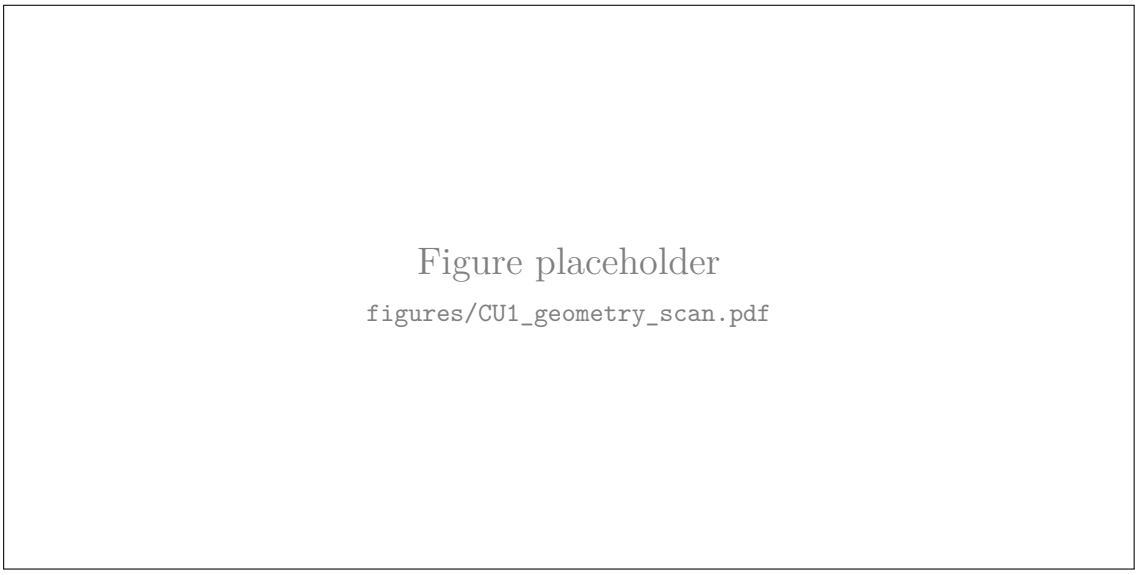


Figure 5.4: CU1 cleanup versus transverse position y_0 and beam height $4\sigma_y$ at constant peak intensity (run `vertical_position_beam_diameter`, earlier configuration with 1.0 MHz switching). To be repeated at constant power.

5.6 Transverse velocity and position

The molecules that reach the detection region have transverse velocities of up to about 1 m/s along the laser beams [2], which shifts the lines by up to 3.7 MHz, more than the whole width of the CU1 optimum (Table 5.1). The transverse velocity v_x was therefore scanned together with the transverse offset y_0, with the Doppler shift of Eq. (4.10): v_x from -1 to $+1$ m/s in steps of 0.1 m/s and y_0 from 0 to 15 mm in steps of 1.5 mm (the problem is symmetric in y_0), at 80 mW, +20.2 MHz and the 4.8 cm beam. Each of the 231 points is one trajectory; the grid was computed for the SPB-related initial state (Sec. 4.8) and for the uniform 12-level state (runs `doppler_cu1_spb` and `doppler_cu1_uniform`, script `scripts/cleanup_spb_doppler.py`, about 5 min each on 8 threads). Two checks: at $v_x = 0$, $y_0 = 0$ the grid reproduces the on-axis values (99.965 % and 99.963 %); and at $v_x = \pm 1$ m/s the remaining population, 0.22, lies between the values of the unmodulated detuning scan at the two shifted detunings, 20.2 ∓ 3.7 MHz (0.17 and 0.20), as expected when half of the passes see each.

Figure 5.5 and Table 5.3 show the result for the SPB-related start.

- **The transverse velocity dominates.** On the axis the cleanup exceeds 99.9 % only for $|v_x| \leq 0.2$ m/s and 99 % only for $|v_x| \leq 0.4$ m/s. It falls to 97.1–97.7 % at 0.5 m/s and to 78 % at 1 m/s. With the sign alternating from pass to pass, a molecule sees the laser at $\delta \pm k_L v_x$, and once $k_L v_x / 2\pi \gtrsim 1.5$ MHz both lie outside the 2 MHz wide 99 % window of Table 5.1.

- **The transverse position matters less.** At $v_x = 0$ the cleanup is 99.945 % at $y_0 = 4.5$ mm, 99.82 % at 9 mm, 99.50 % at 12 mm and 98.54 % at 15 mm, where the intensity has fallen to 46 % of its value on the axis.

- **The map is slightly asymmetric in v_x.** The best point is at $v_x = +0.1\,\mathrm{m/s}$ (99.970 %), and $+v_x$ is better than $-v_x$, because the pass powers are unequal (Sec. 4.7). The first and strongest pass sees $\delta + k_L v_x$, which for $v_x > 0$ moves it towards the resonances of the SPB-related levels, 0.5–0.7 MHz above +20.2 MHz (Sec. 5.9).
- **The uniform 12-level start behaves the same way**, slightly better at large $|v_x|$: 98.0–98.4 % at 0.5 m/s and 84.4 % at 1 m/s on the axis.

Table 5.3: CU1: cleanup efficiency versus transverse velocity at $y_0 = 0$ and $y_0 = 12\,\mathrm{mm}$ (80 mW, +20.2 MHz, SPB-related initial state, run `doppler_cu1_spb`). Two values are for $-|v_x|$ / $+|v_x|$.

$\lvert v_x\rvert$ (m/s)	$y_0 = 0$	$y_0 = 12\,\mathrm{mm}$
0	99.965 %	99.50 %
0.1	99.956 / 99.970 %	99.39 / 99.51 %
0.2	99.929 / 99.959 %	99.12 / 99.37 %
0.3	99.77 / 99.85 %	98.19 / 98.64 %
0.4	99.02 / 99.28 %	95.63 / 96.37 %
0.5	97.13 / 97.65 %	91.35 / 92.28 %
0.7	90.2 / 90.9 %	80.7 / 81.5 %
1.0	78.1 / 78.3 %	66.4 / 66.9 %

5.7 Average over the transverse distribution and choice of detuning

To compare settings by one number, the grids are averaged over an assumed transverse distribution: v_x Gaussian with 1.8 m/s FWHM ($\sigma = 0.76\,\mathrm{m/s}$), truncated at $|v_x| \le 1\,\mathrm{m/s}$ because only these molecules reach the detection region (the cut keeps 81 % of the full Gaussian), and y_0 uniform within $|y_0| \le 15\,\mathrm{mm}$, the coverage of the cylindrical lens [2]. The averages are trapezoidal sums on the grid; the stored grids can be re-weighted with any other distribution. For the comparison of settings a coarser grid was used (v_x in steps of 0.2 m/s, y_0 in steps of 3 mm); on the two full maps it agrees with the fine grid to 0.1 % (93.04 against 93.1 %, 96.06 against 96.15 %).

Table 5.4 lists the results. At the on-axis optimum, +20.2 MHz, the average is only 95.5 % on the axis in y_0 and 93.0 % over the full distribution. Averaged over v_x, the cleanup without EOM has two maxima as a function of detuning (Fig. 5.6a, row $\beta = 0$), at +19.0 MHz (97.4 %) and at +22.0 to +22.5 MHz (97.0–97.2 %), on either side of the on-axis optimum, where the average has a minimum. Over the full distribution +19.0 MHz is the better of the two, 95.2 % against 94.8 %. At these detunings the cleanup on the axis at $v_x = 0$ is worse (96.4 % at +19.0 MHz), but the molecules with $|v_x| = 0.4$–$0.6\,\mathrm{m/s}$, which are more numerous, are cleaned to 99.3–99.8 % instead of 94–99.3 % (Fig. 5.6b).

The distribution above was a first assumption. It has since been replaced by the result of a beam simulation and by the acceptance of the detection laser (Sec. 5.10): v_x Gaussian with $\sigma = 0.65\,\mathrm{m/s}$, y_0 Gaussian with $\sigma = 10.95\,\mathrm{mm}$, counted only within $|v_x| \le 1\,\mathrm{m/s}$ and $|y_0| \le 8.5\,\mathrm{mm}$. The last column of Table 5.4 re-weights the same grids with it. The averages

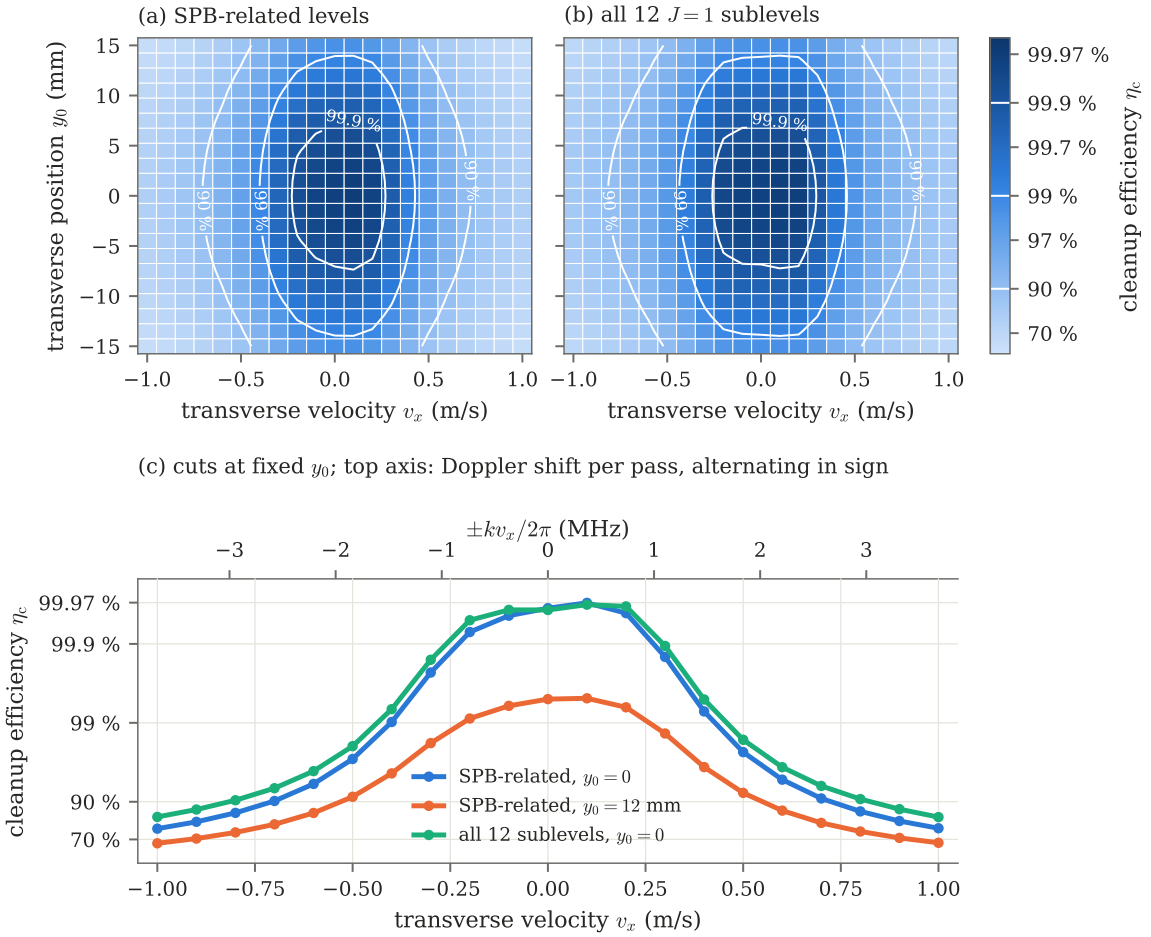


Figure 5.5: CU1 cleanup versus transverse velocity v_x and transverse position y_0 (80 mW, +20.2 MHz, 4.8 cm beam, no EOM). (a) SPB-related initial state, (b) uniform 12-level initial state; logarithmic colour scale, white contours at 90, 99 and 99.9 %; the grid was computed for $y_0 \geq 0$ and mirrored. (c) Cuts at fixed y_0; the top axis gives the Doppler shift per pass, whose sign alternates from pass to pass.

rise by 1 to 2.5 %, but the ranking is unchanged: +19.0 MHz remains the best detuning without EOM, and the EOM gains a further 0.8 %.

5.8 A broadening EOM for CU1

Because the Doppler shift alternates in sign, it cannot be compensated by the laser frequency, only covered by a broader laser spectrum. A phase EOM at 1.6 MHz, available in the laboratory, was tested for this purpose (Sec. 4.7). Its sideband spacing is close to $\Gamma/2\pi$, so the modulated spectrum is nearly continuous, with a width of about $2(\beta + 1) \times 1.6$ MHz, 8 MHz at $\beta = 1.5$. The modulation depth was scanned from 0 to 4 in steps of 0.5, with 1.25 and 1.75 added near the optimum, against the detuning from +16 to +23.5 MHz (part of the plane only, Fig. 5.6a), each point averaged over v_x at $y_0 = 0$; the best settings were then averaged over y_0 as well (script `scripts/cu1_eom_doppler_scan.py`, checkpoints in `tmp/cu1_eom_doppler/`). With $\beta = 0$ the modulated model reproduces the unmodulated

Table 5.4: CU1: cleanup efficiency on the axis ($v_x = 0$, $y_0 = 0$), averaged over v_x at $y_0 = 0$, and averaged over v_x and y_0 with the first assumed distribution (Gaussian v_x, 1.8 m/s FWHM, $|v_x| \leq 1$ m/s; y_0 uniform in $|y_0| \leq 15$ mm). Last column: the same grids averaged with the beam-simulation distribution within the detection acceptance (Sec. 5.10). 80 mW, 4.8 cm beam, SPB-related initial state; coarse grid. The adopted set point is in bold.

			v_x- and y_0-averaged	
setting	on axis	v_x-av., $y_0 = 0$	first assumption	beam sim., acceptance
no EOM, +20.2 MHz	99.965 %	95.50 %	93.04 %	95.46 %
no EOM, +19.0 MHz	96.36 %	97.44 %	95.16 %	97.08 %
no EOM, +22.0 MHz	98.70 %	96.97 %	94.83 %	96.83 %
no EOM, +22.5 MHz	95.48 %	97.18 %	94.72 %	96.73 %
EOM $\beta = 1.0$, +19.0 MHz	98.49 %	98.12 %	95.87 %	97.73 %
EOM $\beta = 1.25$, +19.5 MHz	99.41 %	98.15 %	96.00 %	97.85 %
EOM $\beta = 1.5$, +20.0 MHz	99.45 %	98.17 %	96.06 %	97.89 %
EOM $\beta = 1.75$, +20.0 MHz	99.03 %	98.23 %	96.03 %	97.87 %
EOM $\beta = 2.0$, +20.2 MHz	97.91 %	98.11 %	95.76 %	97.66 %

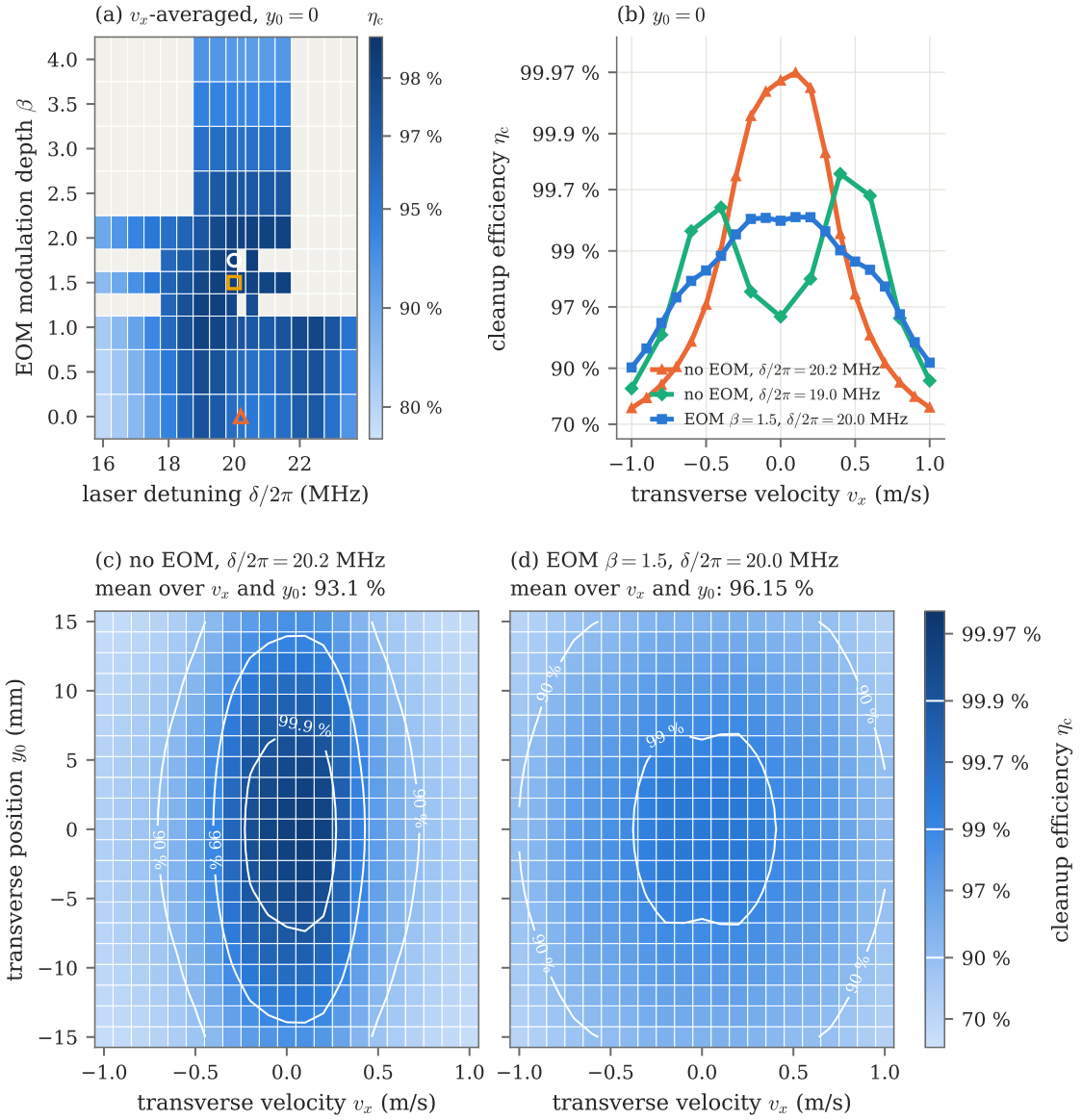
one exactly.

- The v_x-averaged cleanup has a broad maximum of 98.1–98.2 % for $\beta = 1$–2 near +20 MHz (best grid point $\beta = 1.75$, +20.0 MHz, 98.23 %). Beyond $\beta \approx 2$ the power is spread too thinly; at $\beta = 4$ the average drops to about 92 %.
- Over the full distribution the best setting is $\beta = 1.5$ at +20.0 MHz, 96.06 % (96.15 % on the full map, run `doppler_cu1_spb_eom1p6MHz_beta1p5_det20MHz`). The flat-topped velocity response (Fig. 5.6b) costs some efficiency on the axis (99.45 %) but keeps 98.2–99.0 % at $|v_x| = 0.4$–0.6 m/s.
- The gain over the best setting without EOM is 0.9 % (95.16 % to 96.06 %), i.e. the remaining population falls from 4.8 to 3.9 %. What is left is limited by the beam edges in y_0 and by the tails $|v_x| \geq 0.8$ m/s, which need more power rather than more bandwidth.
- In the model the EOM and the polarization switching start with a fixed relative phase. In the experiment they are not synchronised; the effect of the relative phase has not been checked.

Because the gain is small, the EOM is not adopted for now. **The CU1 set point is** +19.0 MHz **without EOM**, which gives 95.2 % over the assumed distribution, against 93.0 % at the on-axis optimum.

5.9 Trajectory at the set point

Figure 5.7 shows a trajectory on the axis ($v_x = 0$, $y_0 = 0$) at +19.0 MHz, computed once for each of the 12 starting sublevels (script `scripts/cu1_trajectories.py`, populations every 0.1 µs, trace constant to 2×10^{-14}). From the SPB-related start 3.6 % remains in $J = 1$, against 0.035 % at +20.2 MHz; from the uniform start 2.5 %. Each molecule scatters 1.49



Averages: v_x Gaussian, FWHM 1.8 m/s, truncated at $|v_x| \le 1$ m/s; y_0 uniform in $|y_0| \le 15$ mm.
Markers in (a): circle best grid point, square setting of (d), triangle present setting (c); grey cells not computed.
80 mW, 4.8 cm beam, 1.6 MHz EOM, 8 SPB-related initial levels; single trajectories.

Figure 5.6: CU1 with a 1.6 MHz phase EOM. (a) Cleanup averaged over v_x at $y_0 = 0$ versus modulation depth and detuning; grey cells were not computed. (b) Cleanup versus v_x at $y_0 = 0$ at the on-axis optimum, at the set point chosen without EOM and at the best EOM setting. (c), (d) Maps over v_x and y_0 without and with the EOM, as in Fig. 5.5, with the average over the distribution. SPB-related initial state, 80 mW, 4.8 cm beam.

photons (1.55 at +20.2 MHz). The removed molecules go to $J = 3$ (47.5 %) and $J = 2$ (46.2 %), with 2.6 % in $J = 4$ and 7×10^{-5} in $J = 0$.

Almost all of the remaining population is in two levels: $|F_1 = \frac{3}{2}, F = 2, m_F = \pm 2\rangle$ and $|F_1 = \frac{1}{2}, F = 0, m_F = 0\rangle$, of which 9.4 and 9.0 % remain when the molecule starts there. Every other sublevel keeps 0.22–0.27 %. The reason is the ground-state structure in the field. The 12 sublevels form two groups 2.4 MHz apart: the eight SPB-related levels, mainly $|m_J| = 1$, lie lower, and the four mainly $m_J = 0$ levels, which include the main ground state, lie higher.

The transitions from the lower group to the excited $|m'_F| = 1$ components are about 2.3 MHz higher in laser frequency (+20.7 to +20.9 MHz against +18.5 MHz). At +19.0 MHz the laser is 0.5 MHz from these components for the upper group but 1.7–1.9 MHz for the lower group. The two levels left behind are in the lower group and couple only weakly to the $|m'_F| = 1$ components: $|M|^2 = 0.005$ for $m_F = \pm 2$ (X polarization) and 0.003 for $F = 0$, which is reached only through its $F = 2$ admixture, against 0.014 to 0.106 for the other levels of the group (X or Z polarization; relative units). With Z polarization the stretched levels reach only $m'_F = \pm 2$, whose resonance lies 17 MHz below the laser. A Doppler shift of about 1.9 MHz ($|v_x| \approx 0.5$ m/s) brings the lower group into resonance in half of the passes, which is why this detuning works better for the velocity distribution than on the axis.

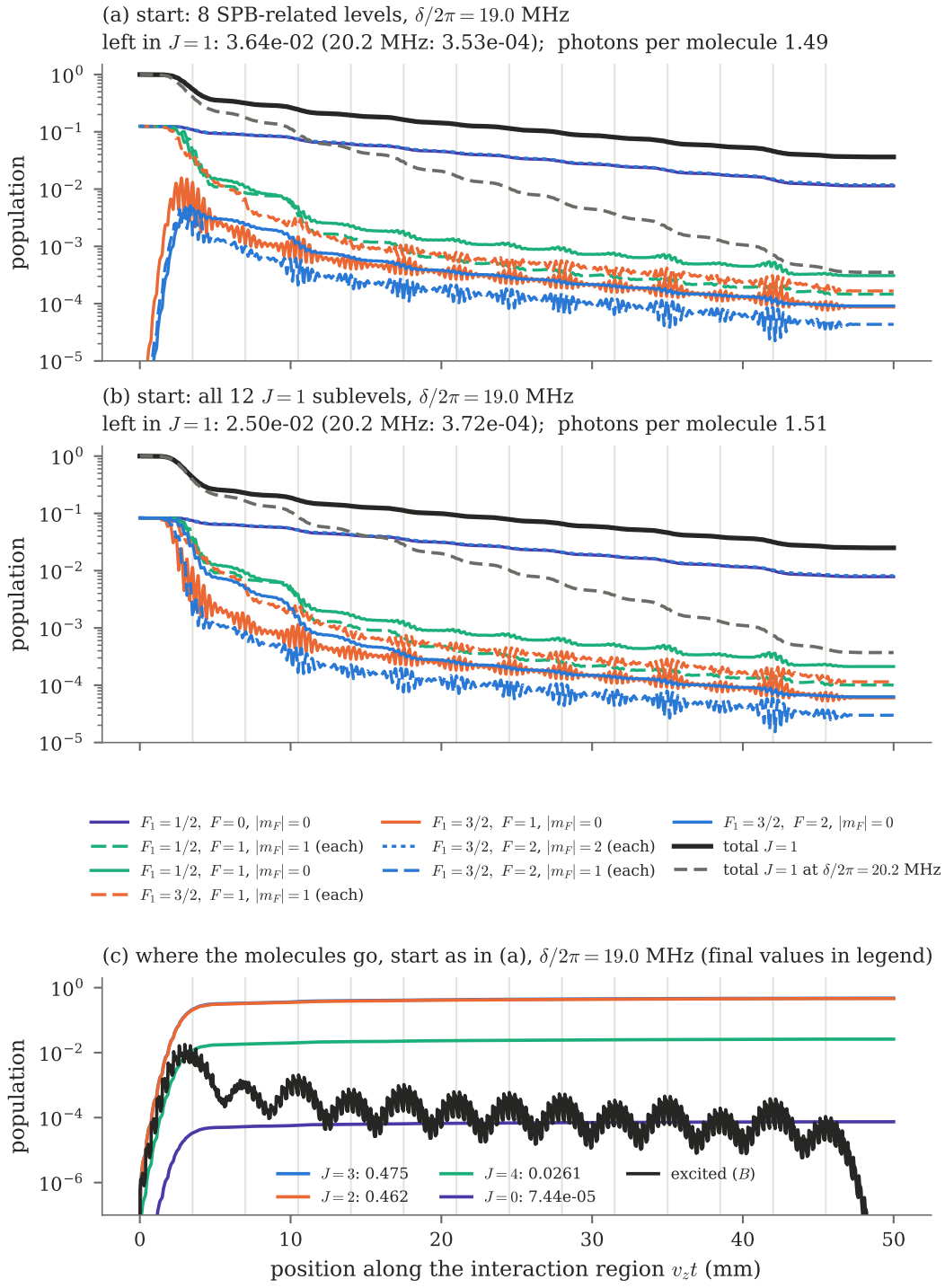
5.10 CU1 at the present beam height and beam distribution

Present conditions. Three inputs have changed since the results above.

- **Beam height.** In the experiment both cleanup beams are now about 3 cm high ($\sigma_y = 7.5$ mm). At the same 80 mW the peak intensity is 1.6 times that of the 4.8 cm beam; the model scales the power normalisation with the height (`build_cu1(sigma_y_m=7.5e-3)`).
- **Beam distribution.** A simulation of the molecular beam with the electrostatic lens at 25 kV gives, at CU1 (lens + 0.5 m), Gaussian transverse positions with $\sigma_x \approx \sigma_y = 10.95$ mm, and transverse velocities with $\sigma_{v_x} \approx \sigma_{v_y} = 0.65$ m/s (1.53 m/s FWHM), the same at both cleanup stages. The two transverse directions are equivalent, so σ_{v_x} is used for the Doppler shift and σ_y for the position along the beam height.
- **Detection acceptance.** The detection laser is 2 cm wide (x) and 1.7 cm high (y). Molecules outside it are not detected, so their cleanup does not matter. The average is therefore taken over $|y_0| \leq 8.5$ mm, and over $|v_x| \leq 1$ m/s as before.

The *mean in acceptance* is the average of the remaining population with Gaussian weights in v_x and y_0 inside this window. The logarithm of the remaining population is interpolated linearly in y_0 between grid points, since 8.5 mm falls between them; on the CU1 map of Sec. 5.6, halving the y_0 resolution changes the mean by 0.015 %. The cut is applied at the cleanup plane, i.e. the transverse drift $v_y L/v$ between the cleanup and the detection region is neglected. For a path length of 2 m it is about 7 mm r.m.s., comparable with the cut, so the population that is eventually detected is not exactly the truncated Gaussian at the cleanup plane; the distribution of the detected molecules at the cleanup planes, taken from the beam simulation, would remove this approximation.

Choice of detuning. With the higher intensity the detuning was scanned again from +17 to +23 MHz in steps of 0.5 MHz, each point averaged over v_x (0.2 m/s steps) and y_0 (0, 3, 6, 9 mm), SPB-related start, no EOM (script `scripts/cu1_3cm_scan.py`, 572 trajectories, about 12 min). Table 5.5 gives the result. The on-axis optimum moves to +20.5 MHz



Single trajectory on the beam axis, $v_x = 0$, $y_0 = 0$; 80 mW, 4.8 cm beam, 13 passes (grey lines: pass centres), 10 % loss per pass, X/Z switching at 0.5 MHz. Populations sampled every 0.1 µs.

Figure 5.7: CU1 at the set point +19.0 MHz on the axis ($v_x = 0$, $y_0 = 0$), 80 mW, no EOM. (a) $J = 1$ sublevel populations from the SPB-related start (one curve per $\pm m_F$ pair, equal by symmetry), with the total and, for comparison, the total at +20.2 MHz. (b) The same from the uniform 12-level start. (c) Populations of the other rotational levels and of the excited state, start as in (a). Grey lines: pass centres.

(99.999 %), but the mean in acceptance is again best at +19.0 MHz (98.66 %); a parabola through the three best points puts the maximum at +19.1 MHz, only 0.05 % higher. A second maximum lies at +22.5 MHz (98.43 %). **The set point +19.0 MHz is kept.**

Table 5.5: CU1 with the 3 cm beam (80 mW, no EOM, SPB-related start): cleanup on the axis, averaged over v_x at $y_0 = 0$, and mean in the detection acceptance ($|v_x| \leq 1$ m/s, $|y_0| \leq 8.5$ mm, beam-simulation weights). The set point is in bold.

$\delta/2\pi$ (MHz)	on axis	v_x-averaged, $y_0 = 0$	mean in acceptance
17.0	87.06 %	92.52 %	90.05 %
17.5	90.27 %	95.22 %	93.28 %
18.0	93.61 %	97.41 %	95.99 %
18.5	96.83 %	98.83 %	97.88 %
19.0	99.01 %	99.34 %	98.66 %
19.5	99.85 %	99.12 %	98.44 %
20.0	99.994 %	98.58 %	97.71 %
20.5	99.999 %	98.10 %	97.07 %
21.0	99.988 %	98.03 %	97.01 %
21.5	99.938 %	98.42 %	97.58 %
22.0	99.73 %	98.96 %	98.29 %
22.5	98.75 %	99.18 %	98.43 %
23.0	96.16 %	98.54 %	97.37 %

Map at the set point. The full map at +19.0 MHz (v_x in steps of 0.1 m/s, y_0 in steps of 1.5 mm, run `doppler_cu1_spb_3cm_det19MHz`) is shown in Fig. 5.8. On the axis the cleanup is 99.01 %; the mean in acceptance is 98.70 %, against 97.08 % for the 4.8 cm beam with the same weights (Table 5.4). The narrower beam helps because inside the acceptance the higher intensity counts more than the loss of coverage at large $|y_0|$. Along $y_0 = 0$ the cleanup is

$|v_x|$ (m/s)	0	0.2	0.3	0.4	0.5	0.7	1.0
η_c at $v_x = -\|v_x\|$ (%)	99.01	99.52	99.85	99.97	99.98	99.67	93.4
η_c at $v_x = +\|v_x\|$ (%)	99.01	99.66	99.92	99.99	99.99	99.83	94.7

so the best molecules are not those at rest but those at $|v_x| \approx 0.5$ m/s.

Why $v_x = 0$ is not the best. The explanation follows from the two groups of ground sublevels of Sec. 5.9.

1. In the field the 12 sublevels of $J = 1$ form two groups 2.4 MHz apart. The four mainly-$m_J = 0$ levels, which include the main ground state, reach the driven $|m'_F| = 1$ components at $\delta/2\pi \approx +18.5$ MHz; the eight SPB-related, mainly-$|m_J| = 1$ levels only at +20.7 to +20.9 MHz. The latter group contains the two most weakly coupled levels, $|F_1 = \frac{3}{2}, F = 2, m_F = \pm 2\rangle$ and $|F_1 = \frac{1}{2}, F = 0\rangle$ ($|M|^2 = 0.005$ and 0.003), which limit the cleanup whenever the laser is off their resonance.

2. At +19.0 MHz and $v_x = 0$ the laser is 0.5 MHz from the upper group but 1.7–1.9 MHz below the SPB-related group in all 13 passes. The two weak levels are then pumped slowly, and 1 % of the SPB-related population remains.

3. A molecule with velocity v_x sees the laser at $\delta + k_L v_x$ in one half of the passes and at $\delta - k_L v_x$ in the other (Sec. 4.7). At $|v_x| = 0.5\,\mathrm{m/s}$, $k_L v_x/2\pi = 1.84\,\mathrm{MHz}$, so in half of the passes the laser appears at $+20.84\,\mathrm{MHz}$, on resonance with the SPB-related group, and the weak levels are emptied. This is where the map has its two maxima.

4. The detuning that is best on the axis, $+20.5\,\mathrm{MHz}$, lies within $0.4\,\mathrm{MHz}$ of the SPB-related group for $v_x = 0$. For a moving molecule the Doppler shift then moves both halves of the passes away from that resonance, and without phase modulation the window is narrow: the mean in acceptance is only 97.1 %. With $\sigma_{v_x} = 0.65\,\mathrm{m/s}$ only 30 % of the molecules have $|v_x| < 0.25\,\mathrm{m/s}$, while 58 % lie between 0.25 and $1\,\mathrm{m/s}$. Placing the laser so that the Doppler-shifted passes are resonant, at the cost of the molecules near rest, is therefore better on average.

5. With the 3 cm beam the loss at $v_x = 0$ is small (99.0 %, against 96.4 % with the 4.8 cm beam), because the higher intensity power-broadens the transitions and partly bridges the $1.8\,\mathrm{MHz}$ gap.

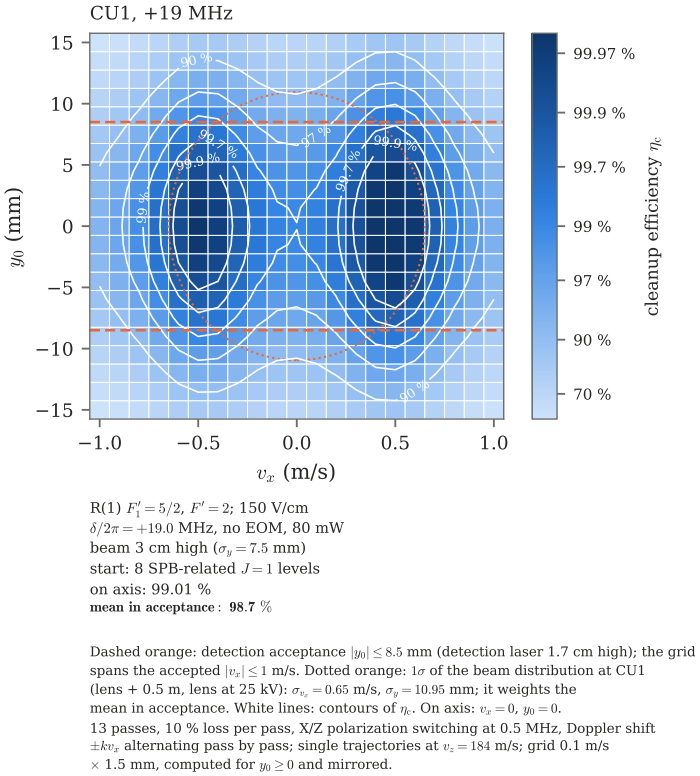


Figure 5.8: CU1 at the present conditions: cleanup efficiency versus transverse velocity and position at $+19.0\,\mathrm{MHz}$, 80 mW, 3 cm beam, no EOM, SPB-related start (run `doppler_cu1_spb_3cm_det19MHz`). Dashed lines: detection acceptance $|y_0| \le 8.5\,\mathrm{mm}$; dotted ellipse: 1σ of the beam distribution at CU1. The conditions and the mean in acceptance are given under the map.

Chapter 6

CU2 results at 200 V/cm

6.1 Operating field

CU2 in the experiment now runs at about 200 V/cm, and this field is used for all CU2 simulations from here on. Earlier simulations at 250 V/cm are collected in Appendix B. They are not combined with 200 V/cm results, because the detuning axes refer to field-dependent main resonances (Sec. 4.5) and because Δ_{main} changes from 219.93 to 211.83 MHz (Eq. (2.1)). At 200 V/cm one-shot state identification is within its validated range and the build issues no ambiguity warning.

6.2 Configuration

Baseline of Table 1.1, 56-level model (Appendix C):

- beam 0 on R(2) $F_1' = 7/2$, $F' = 3$, $P_0 = 80\,\mathrm{mW}$; beam 1 on R(2) $F_1' = 7/2$, $F' = 4$, $P_1 = 24\,\mathrm{mW}$;
- common phase modulation, 17.5 MHz, $\beta = 1.5$;
- X/Z switching at 0.5 MHz in antiphase;
- common multipass, 10 % loss per pass, beam height 2.4 cm;
- frequency difference of the beams $f_{\mathrm{AOM}} = 200 \pm 15\,\mathrm{MHz}$, i.e. $-26.83 \leq (\delta_1 - \delta_0)/2\pi \leq +3.17\,\mathrm{MHz}$ (Table 3.2).

6.3 Single trajectory at the former set point

At the detunings that were optimal at 250 V/cm, $(\delta_0, \delta_1)/2\pi = (+18.5, -6.5\,\mathrm{MHz})$, which at 200 V/cm correspond to $f_{\mathrm{AOM}} = 186.83\,\mathrm{MHz}$, the cleanup is only $\eta_{\mathrm{c}} = 92.87\,\%$, with $N_\gamma = 1.42$. In the 250 V/cm scans the same detunings ($f_{\mathrm{AOM}} = 194.93\,\mathrm{MHz}$ there) gave 99.99 %.

6.4 Laser-frequency scan at fixed AOM frequency

In a first scan both detunings were shifted by a common offset, keeping $(\delta_1 - \delta_0)/2\pi = -25\,\mathrm{MHz}$. This is a scan of the laser frequency at fixed $f_{\mathrm{AOM}} = 186.83\,\mathrm{MHz}$ (Table 6.1, Fig. 6.2). The best point is an offset of $-4\,\mathrm{MHz}$, $(\delta_0, \delta_1)/2\pi = (+14.5, -10.5\,\mathrm{MHz})$, with $\eta_{\mathrm{c}} = 99.33\,\%$. The cleanup changes by several percent between neighbouring points, so the

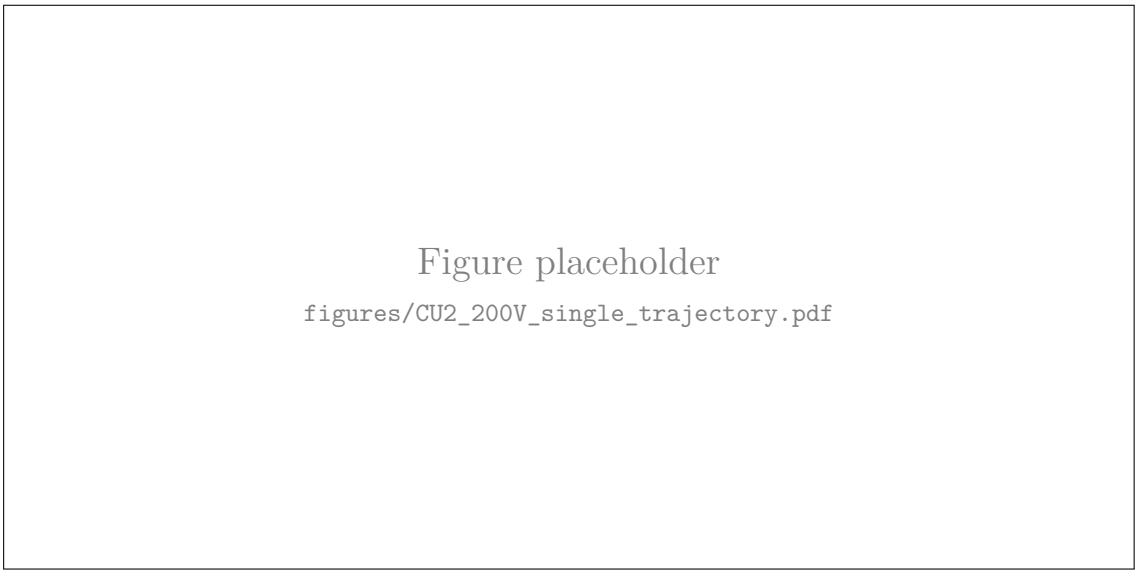


Figure 6.1: CU2 at 200 V/cm, single trajectory: populations of the 20 $J = 2$ sublevels versus distance at $(+18.5, -6.5\,\mathrm{MHz})$, for comparison with the optima in Fig. 6.6.

2 MHz grid is too coarse. For comparison, the CU1 optimum is about 1.6 MHz wide at the 99.5 % level.

Table 6.1: CU2 at 200 V/cm: cleanup efficiency versus a common offset of both detunings at fixed $f_{\mathrm{AOM}} = 186.83\,\mathrm{MHz}$, i.e. $(\delta_1 - \delta_0)/2\pi = -25\,\mathrm{MHz}$ (run `correlated_detuning_shared_offset_200V`, rtol 10^{-6}).

offset (MHz)	$\delta_0/2\pi$ (MHz)	$\delta_1/2\pi$ (MHz)	η_c
−10	8.5	−16.5	96.25 %
−8	10.5	−14.5	95.88 %
−6	12.5	−12.5	97.15 %
−4	14.5	−10.5	99.33 %
−2	16.5	−8.5	97.96 %
0	18.5	−6.5	92.94 %
+2	20.5	−4.5	91.06 %
+4	22.5	−2.5	96.46 %
+6	24.5	−0.5	97.24 %
+8	26.5	1.5	96.14 %

6.5 Why the 250 V/cm set point does not carry over

The field change moves the two manifolds relative to each other. Δ_{main} decreases by 8.1 MHz, and the component positions within each manifold (Table 2.2) move as well. With the AOM frequency of the former optimum, 194.93 MHz, the detuning difference at 200 V/cm would be −16.9 MHz instead of −25 MHz. With the former detuning difference, the AOM must be set to 186.83 MHz. Neither choice reproduces the former alignment of both beams with their manifolds, and a scan of the laser frequency alone, at one AOM frequency, cannot find the new optimum. Laser frequency and AOM frequency must be optimised together.

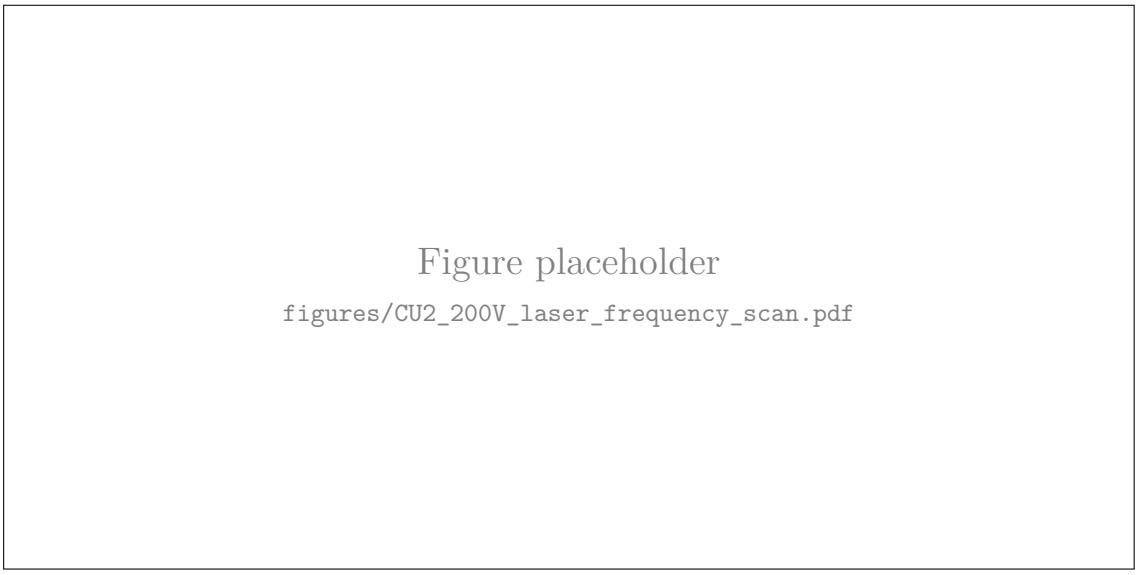


Figure 6.2: CU2 cleanup versus the common detuning offset at 200 V/cm ($f_{\mathrm{AOM}} = 186.83$ MHz, Table 6.1) and, for reference, at 250 V/cm ($f_{\mathrm{AOM}} = 194.93$ MHz, Appendix B). Lower axis: δ_0; upper axis: δ_1.

6.6 Two-dimensional optimisation in laser and AOM frequency

The optimisation uses the experimental coordinates $(\delta_0, f_{\mathrm{AOM}})$, with δ_1 from Eq. (3.1) and $\Delta_{\mathrm{main}} = 211.829$ MHz computed from the model. It was run in three stages with `scripts/cu2_laser_aom_scan.py` (Appendix C), using the scan tolerances of Table 4.3.

1. **Coarse grid**, 2 MHz steps: $\delta_0/2\pi = +5 \ldots + 23$ MHz (about one EOM period around the earlier best point) $\times$ $f_{\mathrm{AOM}} = 185 \ldots 215$ MHz (the full AOM range), 160 points.
2. **Fine grids**, 0.5 MHz steps over ± 2 MHz around the three best separated local maxima of the coarse grid, 219 further points.
3. **Confirmation** of the three best points with rtol 10^{-7}, atol 10^{-9}.

6.6.1 Coarse grid

The cleanup varies between 86.1 % and 99.79 % over the grid (Fig. 6.3). Only 15 of the 160 points exceed 99 %, and 6 exceed 99.5 %. The best points lie in two groups at $\delta_0/2\pi \approx 11$ to 13 MHz, one with $\delta_1/2\pi \approx +8$ MHz ($f_{\mathrm{AOM}} \approx 208$ MHz) and one with $\delta_1/2\pi \approx -10$ MHz ($f_{\mathrm{AOM}} \approx 190$ MHz). The two groups are separated by about one EOM period in δ_1: the $F' = 4$ beam reaches the same components with its carrier or with an adjacent sideband. A third, weaker maximum lies at the edge of the δ_0 range, near $\delta_0/2\pi = 23$ MHz, $f_{\mathrm{AOM}} = 197$ MHz. At $f_{\mathrm{AOM}} = 187$ MHz, next to the line of the earlier laser-frequency scan, the best grid value is 99.48 % at $\delta_0/2\pi = 15$ MHz, consistent with the 99.33 % found there at (14.5, 186.83 MHz). The earlier optimum is therefore only a side maximum.

6.6.2 Fine grids and optimum

Table 6.2 summarises the three fine grids. The overall optimum lies in region 2:

$$\frac{\delta_0}{2\pi} = +11.5\,\mathrm{MHz}, \qquad f_{\mathrm{AOM}} = 191.0\,\mathrm{MHz} \quad \Big(\frac{\delta_1}{2\pi} = -9.33\,\mathrm{MHz}\Big), \qquad \eta_{\mathrm{c}} = 99.85\,\%. \quad (6.1)$$

Region 1 gives almost the same cleanup, 99.80 %, at $(+11.5\,\mathrm{MHz}, 208.5\,\mathrm{MHz})$, $\delta_1/2\pi = +8.17\,\mathrm{MHz}$. Both points lie well inside the AOM range. Compared with the first laser-frequency scan at fixed AOM frequency, the remaining population falls from 0.67 % to 0.15 %.

Table 6.2: CU2 at 200 V/cm: best point of each 0.5 MHz fine grid (runs `laser_AOM_2D_200V_fine_0p5MHz_region1-3`).

region	coarse maximum	best $\delta_0/2\pi$	best f_{AOM}	$\delta_1/2\pi$	η_{c}
1	(11, 209) MHz, 99.79 %	11.5 MHz	208.5 MHz	+8.17 MHz	99.80 %
2	(11, 191) MHz, 99.77 %	11.5 MHz	191.0 MHz	−9.33 MHz	99.85 %
3	(23, 197) MHz, 99.55 %	23.5 MHz	196.5 MHz	+8.17 MHz	99.56 %

6.6.3 Confirmation with tighter tolerances

The three best points, all in region 2, were recomputed with rtol 10^{-7} (Table 6.3). The cleanup changes by 3×10^{-5} at most, so the scan tolerance is adequate at the optimum.

Table 6.3: CU2 at 200 V/cm: best points recomputed with rtol 10^{-7}, atol 10^{-9} (run `laser_AOM_2D_200V_confirm_rtol1e-7`).

$\delta_0/2\pi$	f_{AOM}	η_{c}, rtol 10^{-6}	η_{c}, rtol 10^{-7}
11.5 MHz	191.0 MHz	99.8535 %	99.8508 %
11.0 MHz	191.5 MHz	99.8352 %	99.8319 %
12.0 MHz	190.5 MHz	99.8301 %	99.8275 %

6.6.4 Width of the optimum

Table 6.4 gives the ranges over which the cleanup stays above a given level, along the laser frequency at fixed AOM frequency and along the AOM frequency at fixed laser frequency. A shift along the laser frequency is also what a common shift of both lines produces, for example from a change of the electric field. Both optima are 0.5–1.5 MHz wide at the 99.7 % level and 2–3 MHz wide at 99 %. Region 2 has the higher peak, and region 1 is somewhat wider, in particular along the AOM frequency. The design-study estimate of a 2 MHz line shift from the field variation across the cleanup region (Fig. 3.6) is comparable to the 99 % width, so the field uniformity at 200 V/cm should be checked.

The fine grids also show that both optima are elongated along lines of constant δ_1, $\delta_0 + 2\pi f_{\mathrm{AOM}} = \mathrm{const}$ (Fig. 6.4). Along such a line, i.e. with the $F' = 4$ detuning held fixed, the cleanup in region 2 stays above 99.6 % for $\delta_0/2\pi$ from 10.0 to 13.0 MHz. Moving across it, a change of δ_0 by 0.5 MHz combined with a change of δ_1 by 1 MHz already lowers the cleanup

to 99.4 %. The efficiency is therefore set mainly by the detuning of the $F' = 4$ beam: δ_1 must be held to about ± 0.5 MHz, while δ_0 tolerates about ± 1.5 MHz if δ_1 is kept fixed. Since a drift of the laser frequency shifts both detunings together, the laser frequency is limited by the tighter δ_1 tolerance.

Table 6.4: CU2 at 200 V/cm: ranges (on the 0.5 MHz grid) within which the cleanup exceeds the given level, through the best point of regions 1 and 2.

region	direction	$> 99.7\,\%$	$> 99.5\,\%$	$> 99\,\%$
2	$\delta_0/2\pi$ at $f_{\mathrm{AOM}} = 191.0$ MHz	11.0–12.0 MHz	11.0–12.0 MHz	10.5–12.5 MHz
2	f_{AOM} at $\delta_0/2\pi = 11.5$ MHz	190.5–191.5 MHz	190.0–191.5 MHz	189.5–192.0 MHz
1	$\delta_0/2\pi$ at $f_{\mathrm{AOM}} = 208.5$ MHz	11.5–12.0 MHz	11.0–12.5 MHz	10.5–13.0 MHz
1	f_{AOM} at $\delta_0/2\pi = 11.5$ MHz	208.0–209.5 MHz	208.0–210.0 MHz	207.5–210.5 MHz

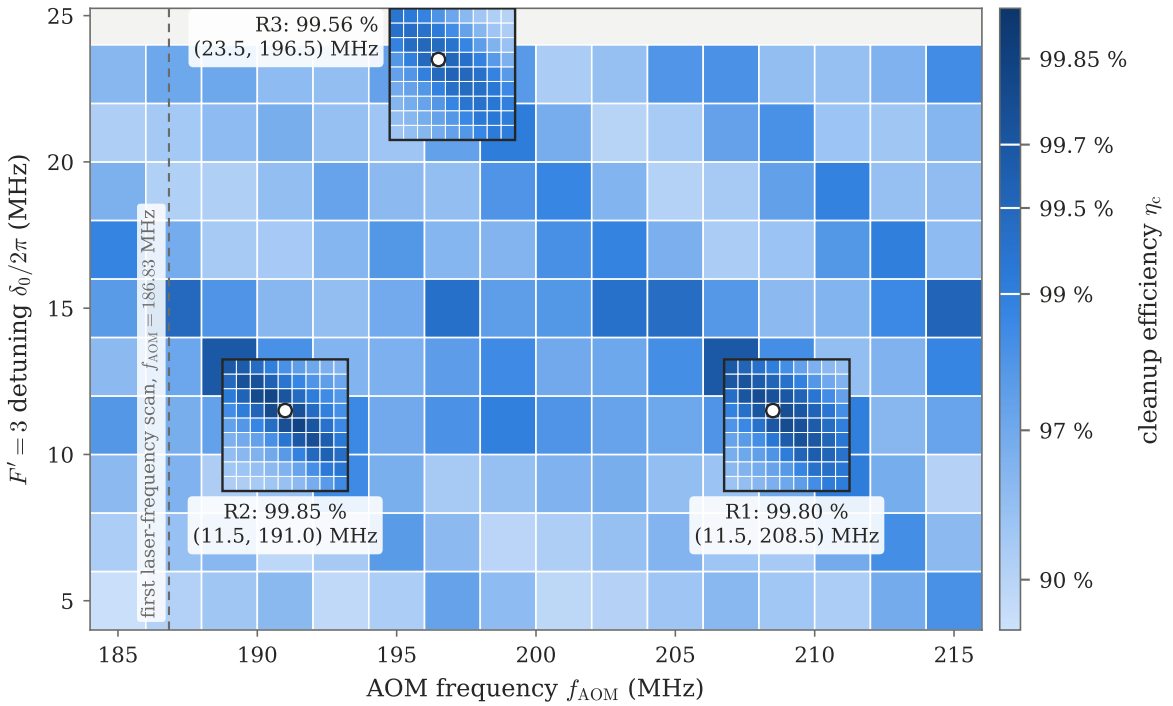


Figure 6.3: CU2 cleanup efficiency at 200 V/cm versus the AOM frequency f_{AOM} (full range 185–215 MHz) and the $F' = 3$ detuning δ_0, i.e. the laser frequency. Colour: remaining $J = 2$ population on a logarithmic scale, labelled in cleanup efficiency; darker is better; white lines on the colour bar mark 99, 99.5 and 99.7 %. Large cells: coarse 2 MHz grid (run `laser_AOM_2D_200V_coarse_2MHz`); outlined boxes: 0.5 MHz fine grids R1–R3, with their best points (circles). Dashed: the line $f_{\mathrm{AOM}} = 186.83$ MHz of the first laser-frequency scan. The grey band above $\delta_0/2\pi = 24$ MHz was not computed outside R3.

6.6.5 Trajectories at the two optima

Single trajectories at the best points of R1 and R2 were computed with rtol 10^{-7}, saving the populations every 0.1 µs. The photon number was taken from the solver's running integral (`output="photon_integral"`) rather than from the sampled excited population. The script is `scripts/cu2_optimum_trajectories.py`. Figures 6.5 and 6.6 show the result, and Table 6.5 lists the final populations.

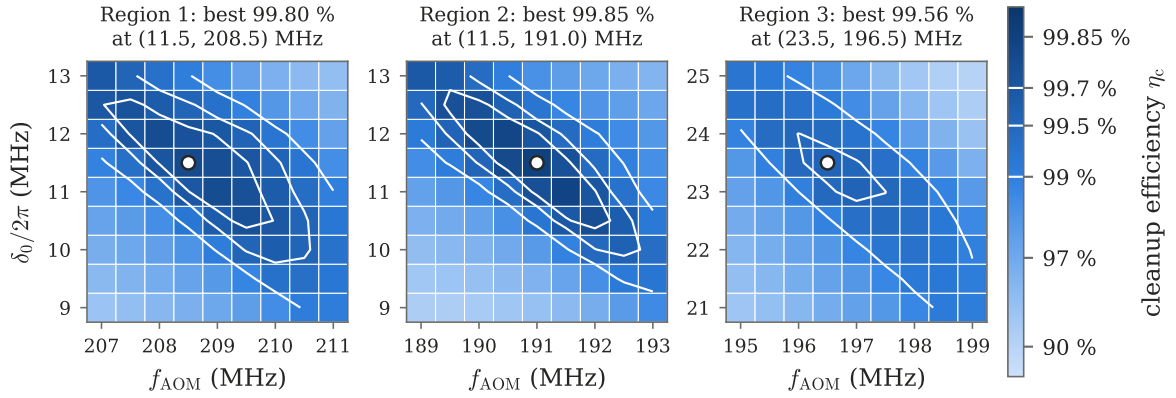


Figure 6.4: CU2 at 200 V/cm: 0.5 MHz fine grids R1–R3 (runs `laser_AOM_2D_200V_fine_0p5MHz_region1-3`), same colour scale as Fig. 6.3. White contours: 99, 99.5 and 99.7 %; circles: best points. The high-efficiency areas run along lines of constant δ_1 (slope -1).

- **Pass by pass.** The first pass removes about 75 % of the $J = 2$ population. Each later pass multiplies what is left by 0.5 to 0.8, following the alternating powers of the multipass passes: strong passes remove more. The population still falls at the last pass, so more passes or more power would improve the cleanup further.
- **Destination.** The removed molecules end almost entirely in $J = 3$ (53–54 %) and $J = 4$ (44–45 %), with 1.7 % in $J = 5$ and 3×10^{-5} in $J = 1$. The excited-state population peaks at 1.4 % in the first pass. Each molecule scatters 1.5 photons on average.
- **Bottleneck.** The remaining population is dominated by the stretched sublevels $|F_1 = \frac{5}{2}, F = 3, m_F = \pm 3\rangle$: 74 % of it in R1 and 56 % in R2. Next come the $F_1 = \frac{3}{2}$, $F = 2$ sublevels, 18 % in R1 and 35 % in R2. R2 cleans the stretched sublevels better, which is why its total is lower.
- **Tolerance check.** With rtol 10^{-7} the R1 optimum gives 99.823 %, against 99.803 % in the scan; the R2 value, 99.851 %, agrees with Table 6.3.

6.6.6 Why both AOM frequencies work

Figure 6.7 shows the excited sublevels addressed by the two beams at 200 V/cm, together with the carrier and sidebands of each beam at the two optima. Its frequency axis is measured from the $F' = 3$ main resonance, so the $F' = 4$ sublevels and the beam-1 lines appear 211.83 MHz higher than on the δ_1 axis used in the text and in Table 6.6. Each sublevel is placed at the coupling-weighted mean of its resonances from the 20 $J = 2$ sublevels; the ground-state hyperfine structure spreads them by at most 1.5 MHz. Which sublevels the light actually

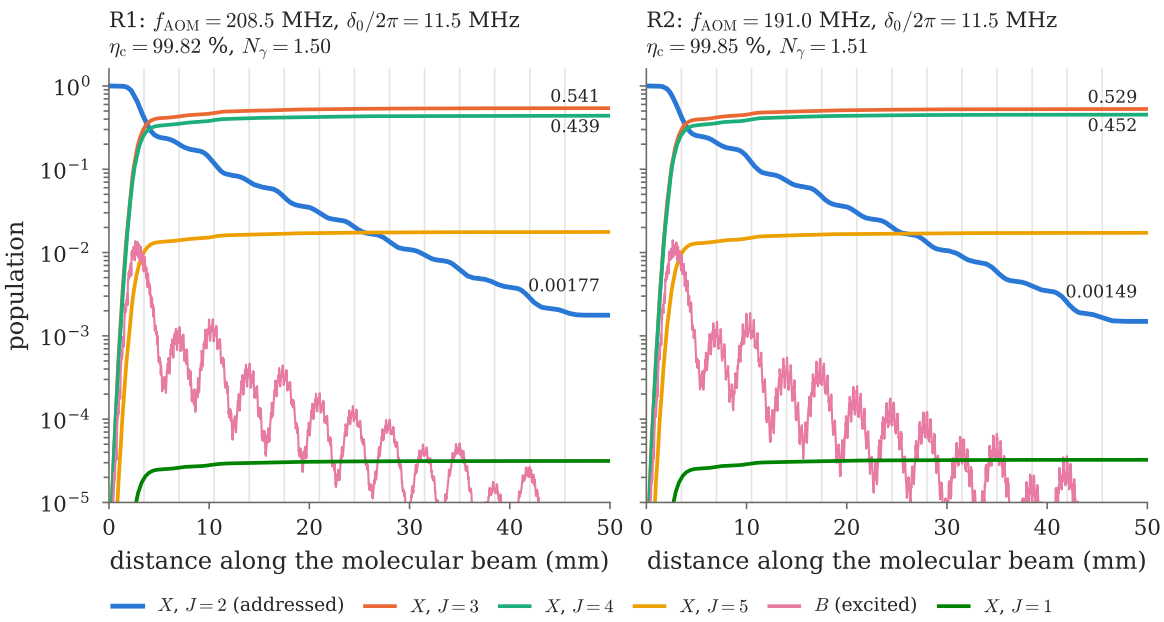


Figure 6.5: CU2 at 200 V/cm: populations of the rotational levels along the interaction region at the optima of R1 ($\delta_1/2\pi = +8.17$ MHz) and R2 ($\delta_1/2\pi = -9.33$ MHz), logarithmic scale. Grey vertical lines: centres of the 13 passes. The excited-state population oscillates with the 0.5 MHz polarization switching and the 17.5 MHz phase modulation; it is sampled every 0.1 µs.

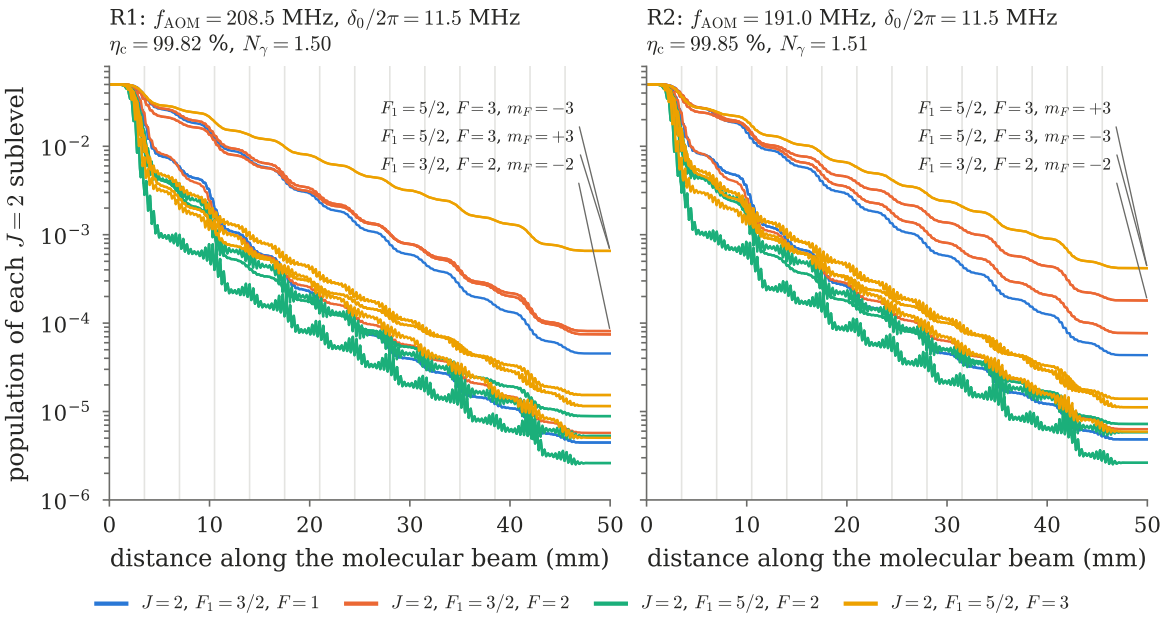


Figure 6.6: CU2 at 200 V/cm: populations of the 20 $J = 2$ sublevels at the optima of R1 and R2, coloured by hyperfine level (F_1, F); the three sublevels with the most population left are labelled. In R1 the two stretched sublevels $m_F = \pm 3$ end at the same value.

Table 6.5: CU2 at 200 V/cm: final populations at the two optima (rtol 10^{-7}), and the photon number per molecule from the running integral.

	R1	R2
$(\delta_0/2\pi,\ f_{\mathrm{AOM}},\ \delta_1/2\pi)$ (MHz)	(11.5, 208.5, +8.17)	(11.5, 191.0, −9.33)
cleanup efficiency η_c	99.823 %	99.851 %
remaining in $J = 2$	1.77×10^{-3}	1.49×10^{-3}
of which $F_1 = \frac{5}{2}$, $F = 3$, $m_F = \pm 3$	74 %	56 %
of which $F_1 = \frac{3}{2}$, $F = 2$	18 %	35 %
$J = 3$	0.541	0.529
$J = 4$	0.439	0.452
$J = 5$	0.0177	0.0173
$J = 1$	3.2×10^{-5}	3.3×10^{-5}
photons per molecule N_γ	1.50	1.51

drives is estimated with a weak-field excitation rate. For sublevel e this is

$$R_e = \sum_{g \in J=2} \sum_p |M_{eg}^{(p)}|^2 \sum_n J_n(\beta)^2 \, \frac{1}{1 + \left[2(\nu_{eg} - \nu_n)/(\Gamma/2\pi)\right]^2}, \tag{6.2}$$

with ν_{eg} the resonance of the pair and ν_n the comb lines; saturation, the polarization switching and coherent effects are not included. The explanation follows from three observations.

1. **The two settings differ by exactly one EOM period.** The AOM frequencies differ by $208.5 - 191.0 = 17.5$ MHz, the EOM frequency, while beam 0 is unchanged. The comb of beam 1, at $\delta_1/2\pi + n \times 17.5$ MHz, therefore occupies the same frequencies in R1 and R2. Only the order n of each line shifts by one: the line that is the carrier in R1 is the first-order sideband in R2, and so on (upper half of Fig. 6.7).

2. **Carrier and first-order sidebands carry almost the same power.** At $\beta = 1.5$ the carrier carries $J_0^2 = 26$ % of the power and each first-order sideband $J_1^2 = 31$ %. The two lines that matter most sit on the strongest $F' = 4$ components in both settings: the line at +8.17 MHz on $m'_F = 0$ (+8.65 MHz, the most strongly coupled $F' = 4$ sublevel), and the line at −9.33 MHz on $m'_F = \pm 2$ (−10.32 MHz). These two components are 19.0 MHz apart, close to the comb spacing, so two adjacent lines can sit on both at once. Whether the carrier or a first-order sideband does the work changes the power on each by only about 16 % (Table 6.6).

3. **The outer lines decide which setting is slightly better.** By Eq. (6.2), beam 1 drives mainly $m'_F = 0$ in both settings, and $m'_F = \pm 2$ at 35 % of that rate in R1 and 25 % in R2. All other $F' = 4$ sublevels stay below 4 %. The bottleneck of Table 6.5, the stretched ground sublevels $|F_1 = \frac{5}{2}, F = 3, m_F = \pm 3\rangle$, is hardly pumped by beam 0: its rate through $F' = 3$ is 40 to 50 times lower than through $F' = 4$. In $F' = 4$ it couples mainly to $m'_F = \pm 4$ (−31.41 MHz) and $m'_F = \pm 3$ (−20.44 MHz), which lie far from the two main lines. R2 has a first-order sideband (31 %) at −26.83 MHz, 4.6 MHz from the strongly coupled $m'_F = \pm 4$; R1 has only a second-order sideband there (5 %). The stretched sublevels are therefore pumped 15 % faster in R2. R1 in turn has 31 % instead

of 26 % on $m'_F = \pm 2$, and pumps the $F_1 = \frac{3}{2}$, $F = 2$ sublevels 11 % faster. Both trends agree with the composition of the remaining population: the stretched sublevels are 56 % of it in R2 against 74 % in R1, and $F_1 = \frac{3}{2}$, $F = 2$ is 35 % against 18 %. Since the stretched sublevels are the main bottleneck, R2 ends with less population left, 99.85 % against 99.82 %.

Because the comb repeats every 17.5 MHz, the optimum repeats every 17.5 MHz in f_{AOM}. Within the AOM range only these two copies exist: the next ones, at 173.5 and 226 MHz, lie outside 185–215 MHz. Beam 0 is the same in both settings. Its first-order sideband sits on the $F' = 3$, $m'_F = \pm 1$ components (+28.41 MHz), and its carrier lies 3 MHz below the $m'_F = \pm 2$ components.

Table 6.6: CU2 at 200 V/cm: comb lines of beam 1 near the $F' = 4$ sublevels, with their order n and power fraction $J_n(1.5)^2$ in R1 and R2. Positions on the δ_1 axis.

line (MHz)	nearest $F' = 4$ sublevel(s)	R1: n, power	R2: n, power
+8.17	$m'_F = 0$, $P' = -$ at +8.65 (strongest)	0, 26 %	+1, 31 %
−9.33	$m'_F = \pm 2$, $P' = -$ at −10.32	−1, 31 %	0, 26 %
+25.67	$m'_F = \pm 1$, $P' = +$ at +22.50	+1, 31 %	+2, 5 %
−26.83	$m'_F = \pm 3$ at −20.83, ± 4 at −31.41, $P' = -$	−2, 5 %	−1, 31 %
+43.17	$m'_F = \pm 3$, $P' = +$ at +44.43	+2, 5 %	+3, 0.4 %

6.7 Transverse velocity and position at the present beam height

Setup. The two optima were evaluated over the transverse velocity and position under the present conditions of Sec. 5.10: beam height 3 cm ($\sigma_y = 7.5$ mm) at the same 80 + 24 mW, i.e. 0.8 times the peak intensity of the 2.4 cm beam used for the optimisation (`build(sigma_y_m=7.5e-3)`). The two beams run through the same multipass in the same direction, so the Doppler shift, Eq. (4.10), is common to both: in the $(\delta_0, f_{\mathrm{AOM}})$ plane it moves the operating point along the laser frequency at fixed AOM frequency, in alternate directions in alternate passes. A coarse grid was computed, v_x from −1 to +1 m/s in steps of 0.2 m/s and y_0 from 0 to 15 mm in steps of 3 mm (66 trajectories, about 50 min per run), for R1 and R2 and for two initial states, all 20 $J = 2$ sublevels and the four SPB source levels ($J = 2$, $m_J = 0$, lens weights 2/3 for the singlet and 1/9 for each triplet level) (script `scripts/cu2_doppler_scan.py`, runs `doppler_cu2_{R1,R2}_{uniform,spb}_3cm_coarse`). The means use the beam-simulation distribution at CU2 (lens + 1.0 m): $\sigma_{v_x} = 0.65$ m/s and $\sigma_y = 10.35$ mm, within the detection acceptance $|v_x| \leq 1$ m/s, $|y_0| \leq 8.5$ mm.

Results. Table 6.7 and Fig. 6.8 summarise the four runs.

- **The transverse position dominates.** With $\sigma_y = 7.5$ mm the intensity falls to 73, 49, 28 and 14 % of its axial value at $y_0 = 6$, 9, 12 and 15 mm. From the uniform start

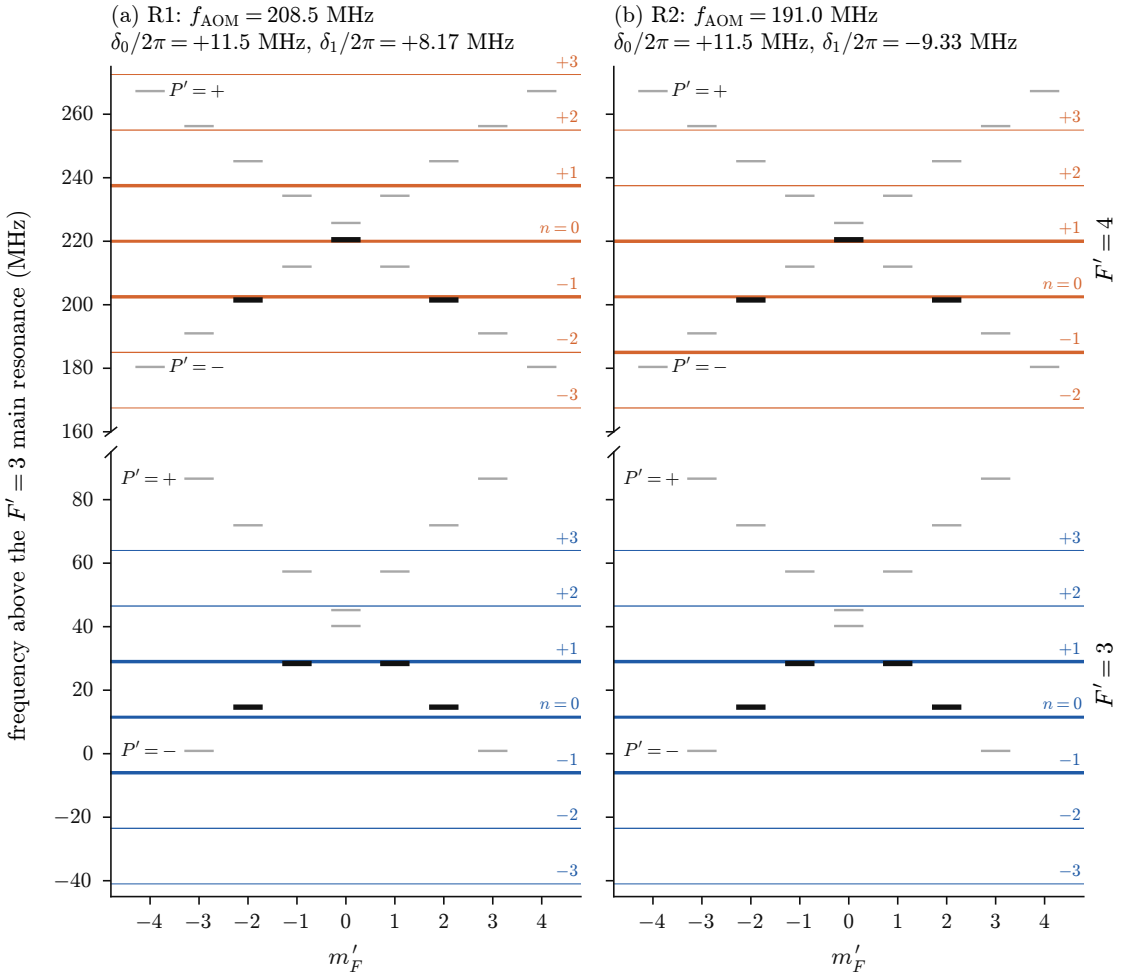


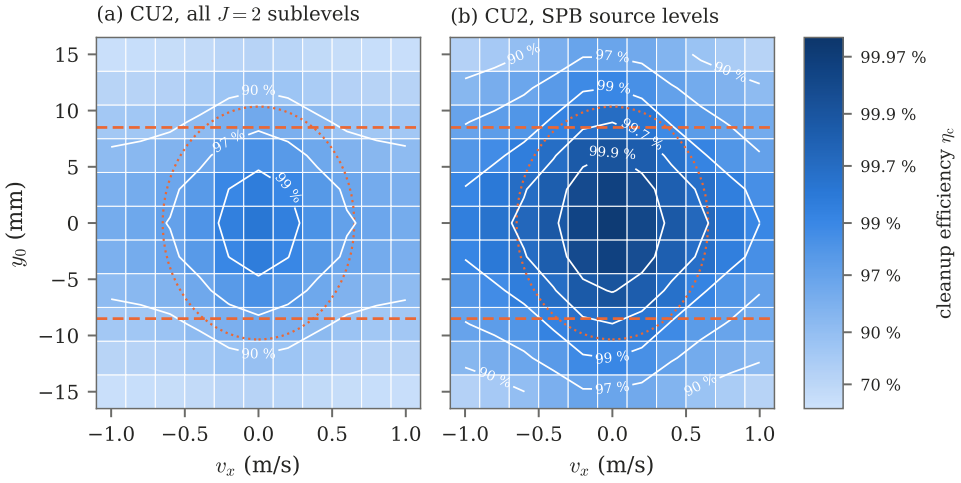
Figure 6.7: CU2 at 200 V/cm: excited sublevels of B, $J' = 3$, $F'_1 = 7/2$ addressed from X, $J = 2$, and the laser combs at the optima of (a) R1 and (b) R2. The frequency axis is broken between $F' = 3$ (bottom) and $F' = 4$ (top) and gives the transition frequency above the $F' = 3$ main resonance; the $F' = 4$ main resonance lies at 211.83 MHz. Each dash is one excited sublevel at the coupling-weighted mean of its resonances from the 20 $J = 2$ sublevels (which span at most 1.5 MHz). The two Λ-doublet components with the same m'_F are aligned vertically and form the $P' = -$ and $P' = +$ branches. Sublevels driven by the light, with an excitation rate, Eq. (6.2), of at least 5 % of the most strongly driven sublevel of the same beam, are shown in bold. Coloured lines: carrier ($n = 0$) and phase-modulation sidebands n of beam 0 (blue, $F' = 3$) and beam 1 (orange, $F' = 4$), with line width proportional to the power fraction $J_n(1.5)^2$ (0.26, 0.31, 0.05 and 0.004 for $|n| = 0$–3). Beam 0 is the same in (a) and (b). The beam-1 lines lie at the same frequencies in both columns, and only their orders differ by one. Script: `scripts/cu2_level_scheme.py`.

Table 6.7: CU2 at 200 V/cm with the 3 cm beam: cleanup on the axis, averaged over v_x at $y_0 = 0$, and mean in the detection acceptance; for comparison, the mean with the y_0 range extended to 15 mm (coarse grid, beam-simulation weights). The recommended setting is in bold.

f_{AOM}	start	on axis	v_x-av., $y_0 = 0$	**in acceptance**	$\lvert y_0\rvert \le 15$ mm
R2, 191.0 MHz	all 20 sublevels	99.59 %	97.72 %	96.10 %	91.84 %
R1, 208.5 MHz	all 20 sublevels	99.52 %	98.16 %	96.66 %	92.51 %
R2, 191.0 MHz	SPB source levels	99.975 %	99.67 %	99.35 %	98.42 %
R1, 208.5 MHz	SPB source levels	99.974 %	99.81 %	99.57 %	98.79 %

at $v_x = 0$ the cleanup falls accordingly from 99.5–99.6 % on the axis to 98.6 % at 6 mm, 96.0 % at 9 mm and 88.7–89.0 % at 12 mm. The detection acceptance cuts this tail: over $|y_0| \leq 15$ mm the mean would be about 4 % lower for the uniform start.

- **The transverse velocity matters much less than for CU1.** From the uniform start on the axis the cleanup is still 97.8–98.4 % at $|v_x| = 0.4$ m/s and 94.7–95.5 % at 1 m/s, against 78 % at 1 m/s for CU1 at its on-axis optimum (Table 5.3). Each CU2 beam carries a comb of five significant lines and the two beams address the 32 sublevels of $F' = 3$ and $F' = 4$ (both Λ-doublet components), so a shift of a few MHz detunes some lines from their components while others come closer, and the cleanup degrades gradually instead of leaving a narrow resonance.
- **R1 is more robust than R2.** On the axis R2 is marginally better, but over the distribution R1 wins for both starts (96.66 against 96.10 %, 99.57 against 99.35 %); its cleanup falls more slowly with $|v_x|$. With transverse motion included, **R1 ($f_{\mathrm{AOM}} =$ 208.5 MHz) is the recommended CU2 setting.**
- **The SPB source levels are cleaned much better than the average sublevel** (99.6 % against 96.7 % in acceptance), as on the axis (Sec. 6.6.5): they couple strongly to $F' = 4$, while the uniform start contains the slowly pumped stretched levels, which dominate where the intensity is low. Which of the two numbers applies depends on how the $J = 2$ population entering CU2 is divided between the molecules SPB did not transfer and the molecules that CU1 pumped into $J = 2$.
- The lower peak intensity of the 3 cm beam reduces the on-axis cleanup from the uniform start from 99.85 % (2.4 cm) to 99.59 % at R2. R1 and R2 were optimised on the axis for the 2.4 cm beam; a re-optimisation of $(\delta_0, f_{\mathrm{AOM}})$ for the 3 cm beam and the transverse distribution has not been done.



R(2) $F_1'=7/2$, $F'=3+F'=4$; 200 V/cm
setting R1: $f_{\mathrm{AOM}}=208.5$ MHz,
$\delta_0/2\pi=+11.5$, $\delta_1/2\pi=+8.17$ MHz
80 + 24 mW; EOM 17.5 MHz, $\beta=1.5$
beam 3 cm high ($\sigma_y=7.5$ mm)
start: all 20 $J=2$ sublevels
on axis: 99.52 %
mean in acceptance : 96.7 %

R(2) $F_1'=7/2$, $F'=3+F'=4$; 200 V/cm
setting R1: $f_{\mathrm{AOM}}=208.5$ MHz,
$\delta_0/2\pi=+11.5$, $\delta_1/2\pi=+8.17$ MHz
80 + 24 mW; EOM 17.5 MHz, $\beta=1.5$
beam 3 cm high ($\sigma_y=7.5$ mm)
start: 4 $J=2$, $m_J=0$ levels
(lens weights 2/3, 1/9, 1/9, 1/9)
on axis: 99.97 %
mean in acceptance : 99.6 %

Dashed orange: detection acceptance $|y_0|\le 8.5$ mm (detection laser 1.7 cm high); the grid spans the accepted $|v_x|\le 1$ m/s. Dotted orange: 1σ of the beam distribution at CU2 (lens + 1.0 m, lens at 25 kV): $\sigma_{v_x}=0.65$ m/s, $\sigma_y=10.35$ mm; it weights the mean in acceptance. White lines: contours of η_c. On axis: $v_x=0$, $y_0=0$.
13 passes, 10 % loss per pass, X/Z polarization switching at 0.5 MHz, Doppler shift $\pm kv_x$ alternating pass by pass; single trajectories at $v_z=184$ m/s; grid 0.2 m/s × 3 mm, computed for $y_0\ge 0$ and mirrored.

Figure 6.8: CU2 at the present conditions, setting R1 ($f_{\mathrm{AOM}} = 208.5\,\mathrm{MHz}$, $\delta_0/2\pi = +11.5\,\mathrm{MHz}$), 3 cm beam, 80 + 24 mW: cleanup versus transverse velocity and position from (a) all 20 $J = 2$ sublevels and (b) the four SPB source levels (coarse grid, mirrored in y_0). Dashed lines: detection acceptance; dotted ellipse: 1σ of the beam distribution at CU2. The conditions and the mean in acceptance are given under each map.

Chapter 7

Numerical checks

- **Trace.** $\mathrm{Tr}\,\rho$ stays one to better than 10^{-14} in every model: no population leaves the modelled manifold.
- **Reproducibility.** The CU1 coarse and fine detuning scans, run separately, agree exactly at their four common points.
- **Tolerance.** At 200 V/cm and $(+18.5, -6.5\,\mathrm{MHz})$, a point with much population left, the single trajectory with rtol 10^{-7} leaves 0.07125 in $J = 2$ and the scan with rtol 10^{-6} leaves 0.07062, a difference of 6×10^{-4}. At the optimum of the two-dimensional scan the difference is only 3×10^{-5} (Table 6.3), so the scan tolerance is adequate where it matters.
- **Λ-doublet partner.** Effect on the CU1 cleanup: 6×10^{-6}.
- **State identification.** No ambiguity warning at 150 and 200 V/cm; one at 250 V/cm (two states, margin 0.012).
- **Cross-coupling.** All neglected couplings between the two CU2 beams and the other excited level are detuned by more than 50 MHz at every scanned point, the minimum being at $\delta_0/2\pi = 5\,\mathrm{MHz}$, $f_{\mathrm{AOM}} = 185\,\mathrm{MHz}$, and by 63 MHz at the optimum (Sec. 4.5).

Still to be done: convergence in $J_{\max}$ at 200 V/cm for the eigenvectors (not only the energies). Photon numbers from the running integral are available for the two CU2 optima (Sec. 6.6.5); the other photon numbers in this report use the trapezoidal rule on the 1 µs output grid.

Chapter 8

Conclusions and outlook

Table 8.1: Main results (baseline of Table 1.1).

quantity	value
CU1: cleanup at 80 mW, +20.2 MHz, 150 V/cm	99.96 %
CU1: detuning window for $\eta_c > 99.9\,\%$	0.6 MHz
CU1: cleanup at 110 mW (grid point +20.8 MHz)	99.955 %
CU1: effect of the Λ-doublet partner	6×10^{-6}
CU1: $\lvert v_x\rvert$ for $\eta_c > 99.9\,\%$ / $> 99\,\%$ (on axis, +20.2 MHz)	0.2 / 0.4 m/s
CU1: average over v_x (1.8 m/s FWHM) and $\lvert y_0\rvert \leq 15$ mm, +20.2 MHz	93.0 %
CU1: same, at the set point +19.0 MHz (no EOM)	95.2 %
CU1: same, with a 1.6 MHz EOM, $\beta = 1.5$, +20.0 MHz (not adopted)	96.1 %
CU2: $F' = 4$ minus $F' = 3$ main resonance at 200 / 250 V/cm	211.83 / 219.93 MHz
CU2, 200 V/cm: reachable $(\delta_1 - \delta_0)/2\pi$ for $f_{\mathrm{AOM}} = 200 \pm 15$ MHz	−26.83 to +3.17 MHz
CU2, 200 V/cm: cleanup at the former detunings (+18.5, −6.5 MHz)	92.9 %
CU2, 200 V/cm: best at $f_{\mathrm{AOM}} = 186.83$ MHz (laser-frequency scan)	99.33 %
CU2, 200 V/cm: optimum $\delta_0/2\pi = +11.5$ MHz, $f_{\mathrm{AOM}} = 191.0$ MHz (rtol 10^{-7})	99.85 %
CU2, 200 V/cm: second optimum $\delta_0/2\pi = +11.5$ MHz, $f_{\mathrm{AOM}} = 208.5$ MHz	99.80 %
CU2, 200 V/cm: width of the optimum at 99 % (laser / AOM frequency)	2 / 2.5 MHz
CU2 reference at 250 V/cm (Appendix B)	99.991 %
present conditions: 3 cm *beams, beam simulation at* 25 kV*, detection acceptance*	
CU1, +19.0 MHz, no EOM: mean in acceptance / on axis	98.7 % / 99.0 %
CU1: on-axis optimum (3 cm)	+20.5 MHz, 99.999 %
CU2, R1 ($f_{\mathrm{AOM}} = 208.5$ MHz), all 20 sublevels: mean in acceptance / on axis	96.7 % / 99.5 %
CU2, R1, SPB source levels: mean in acceptance / on axis	99.6 % / 99.97 %
CU2, R2 ($f_{\mathrm{AOM}} = 191.0$ MHz), all 20 sublevels / SPB source levels	96.1 % / 99.35 %

On the beam axis CU1 removes 99.96 % of the $J = 1$ population with a single 80 mW beam and no phase modulation, as the design study anticipated. The optimum is narrow (0.6 MHz at the 99.9 % level), and this makes CU1 sensitive to the transverse velocity: in the multipass the Doppler shift alternates in sign from pass to pass, so it cannot be compensated by the laser frequency, and the cleanup falls below 99 % for $\lvert v_x\rvert > 0.4$ m/s. For an assumed Gaussian distribution of 1.8 m/s FWHM (cut at 1 m/s) and a uniform transverse position within 15 mm, the average is 93.0 % at the on-axis optimum and 95.2 % at +19.0 MHz, which is adopted as the set point. A 1.6 MHz phase EOM gains another 0.9 % (96.1 %) and is not adopted for now. The remaining population is then limited by the beam edges in y_0 and by the tails of the velocity distribution, i.e. by the power.

Under the present conditions, a 3 cm beam, the transverse distribution of the beam simulation and the acceptance of the detection laser, CU1 at +19.0 MHz removes 98.7 % of the

$J = 1$ population that is eventually detected (Sec. 5.10); +19.0 MHz remains the best detuning, although the on-axis optimum has moved to +20.5 MHz. At this set point the molecules at rest are cleaned less well (99.0 %) than those at $|v_x| \approx 0.5$ m/s (99.99 %): the eight SPB-related sublevels lie 2.4 MHz below the other four, and only a Doppler shift of about 1.8 MHz brings them into resonance, in half of the passes. Since most molecules move transversely, this trade is favourable on average.

For CU2 the change of field from 250 to 200 V/cm invalidates the former set point, because the relative position of the $F' = 3$ and $F' = 4$ manifolds moves by 8 MHz. Since the two beams are coupled by the AOM, the new optimum had to be found in laser frequency and AOM frequency together, within $f_{\mathrm{AOM}} = 200 \pm 15$ MHz. The two-dimensional optimisation finds 99.85 % at $\delta_0/2\pi = +11.5$ MHz, $f_{\mathrm{AOM}} = 191.0$ MHz, confirmed with tighter tolerances, and a nearly equivalent optimum (99.80 %) at $f_{\mathrm{AOM}} = 208.5$ MHz, exactly one EOM period higher in f_{AOM}. Both work because the comb of the $F' = 4$ beam then lies at the same frequencies, with the roles of carrier and first-order sideband exchanged, and these carry almost the same power at $\beta = 1.5$ (Sec. 6.6.6). The remaining $J = 2$ population is 0.15 %, against 0.67 % for the best point of the first laser-frequency scan. Both optima are 2–3 MHz wide at the 99 % level.

With the present 3 cm beam and the transverse distribution (Sec. 6.7), the second optimum, $f_{\mathrm{AOM}} = 208.5$ MHz, is the more robust one and is recommended: it removes 96.7 % of a uniformly populated $J = 2$ level and 99.6 % of the population in the SPB source levels within the detection acceptance. For CU2 the transverse position matters more than the transverse velocity: the comb of the two phase-modulated beams tolerates Doppler shifts of a few MHz, but the intensity at the edge of the acceptance is only about 50 % of the axial value.

Next steps:

1. Use the beam simulation to give the transverse distribution of the molecules that reach the detection laser, at the CU1 and CU2 planes. This removes the approximation of applying the detection cut at the cleanup planes; the stored grids only need to be re-weighted.

2. CU2: re-optimise $(\delta_0, f_{\mathrm{AOM}})$ for the 3 cm beam with the transverse average, starting from R1, and scan the power split between the two beams. The stretched sublevels $|F_1 = \frac{5}{2}, F = 3, m_F = \pm 3\rangle$, pumped almost only by the 24 mW $F' = 4$ beam, limit the cleanup at low intensity (Sec. 6.6.5).

3. Sensitivity of the CU2 optimum to the electric field (e.g. 198 and 202 V/cm), which requires rebuilding the model, and the field uniformity across the cleanup region at 200 V/cm.

4. Repeat the geometry scans at constant power (normalisation $\propto 1/\sigma_y$); both builders now support it (`sigma_y_m`).

5. Correct the stored metadata of the CU1 run `detuning_withoutP-_-20to_100MHz`, which records `retain_opposite_parity_levels=True` although it was run with the 21-level model.

Appendix A

Model parameters

Table A.1: Parameters of the simulation.

symbol	quantity	value
fields and molecule		
E	electric field along z	150 V/cm (CU1), 200 V/cm (CU2)
B	magnetic field along z	1×10^{-3} G
Γ	B-state decay rate	$1.0101 \times 10^{7}\,\mathrm{s}^{-1}$ ($\Gamma/2\pi = 1.608$ MHz)
D_{XB}	X–B transition dipole moment	$2.667\,550\,6 \times 10^{-30}$ C m
$J_{\max}$	rotational truncation of X and B	4 (CU1), 5 (CU2)
trajectory		
v	forward velocity	184 m/s
v_x	transverse velocity	$-1 \ldots +1$ m/s scanned
$k_L v_x/2\pi$	Doppler shift, sign alternating per pass	3.68 MHz per m/s
y_0	transverse offset	0 . . . 15 mm scanned (mirrored)
σ_{v_x}	transverse velocity spread (beam sim., 25 kV)	0.65 m/s, both stages
σ_y	beam size (beam sim., 25 kV)	10.95 mm (CU1), 10.35 mm (CU2)
	detection acceptance	$\lvert v_x\rvert \le 1$ m/s, $\lvert y_0\rvert \le 8.5$ mm
	first assumption (Sec. 5.7)	v_x: 1.8 m/s FWHM; y_0 uniform in 15 mm
L_{IR}, T	length and duration of the interaction region	5 cm, 271.7 µs
laser beam		
	number of passes	13
s_n	pass centres	$n \times 3.5$ mm, $n = 1 \ldots 13$
L	loss per pass	10 %
σ_s	Gaussian width along the beam	0.75 mm (3 mm full 4σ)
σ_y	Gaussian width transverse	12 mm (CU1, 4.8 cm), 6 mm (CU2, 2.4 cm)
	present beam height, both stages	3 cm ($\sigma_y = 7.5$ mm), constant power
$\omega_{\mathrm{p}}/2\pi$	polarization switching frequency	0.5 MHz, square wave, X/Z
CU1		
	transition	R(1) $F_1' = 5/2$, $F' = 2$
P	laser power	80 mW (scans 10–110 mW)
$\delta/2\pi$	detuning: set point / on-axis optimum	+19.0 MHz / +20.2 MHz
	optional phase EOM (not adopted)	1.6 MHz; best $\beta = 1.5$ at +20.0 MHz
m	main-pair matrix element	0.1004
CU2		
	transitions	R(2) $F_1' = 7/2$, $F' = 3$ and $F' = 4$
P_0, P_1	beam powers	80 mW, 24 mW
$\omega_{\mathrm{m}}/2\pi$, β	phase modulation, common to both beams	17.5 MHz, 1.5
f_{AOM}	AOM frequency (beam separation)	200 ± 15 MHz
Δ_{main}	separation of the main resonances	211.83 MHz (200 V/cm)
$\delta_0/2\pi$, f_{AOM}	optimum (laser, AOM)	+11.5 MHz, 191.0 MHz ($\delta_1/2\pi = -9.33$ MHz)
m_0, m_1	main-pair matrix elements	0.0948, 0.2286
integration		
	method	`dopri5`, Rust, `expanded_sparse`, `decomposed`
	initial step, tolerances	2 ns; rtol 10^{-7}, atol 10^{-9} (CU2 scans: 10^{-6}, 10^{-8})

Appendix B

Reference results at 250 V/cm

These CU2 results were obtained before the operating field was changed to 200 V/cm. They use the same 56-level two-transition model and the same baseline beam parameters, and are kept as a reference only.

- Laser-frequency scan at fixed $f_{\mathrm{AOM}} = 194.93\,\mathrm{MHz}$ ($(\delta_1 - \delta_0)/2\pi = -25\,\mathrm{MHz}$), combined from three runs covering $\delta_0/2\pi = -58.5$ to $+141.5\,\mathrm{MHz}$ in 1 MHz steps (one gap, 100.5 to 102.5 MHz): best $\eta_{\mathrm{c}} = 99.991\,\%$ at $(+18.5, -6.5\,\mathrm{MHz})$ (Fig. B.1). Along this direction the cleanup falls to 99.94 % at $-1\,\mathrm{MHz}$ and 99.83 % at $+1\,\mathrm{MHz}$ from the optimum, and to 99.03 % and 98.68 % at $\mp 2\,\mathrm{MHz}$.
- 3×3 grid around this point (Table B.1).

Table B.1: CU2 at 250 V/cm: cleanup efficiency (%) on a 3×3 grid (run `Fprime3_Fprime4_detuning`).

	$\delta_1/2\pi = -7.5\,\mathrm{MHz}$	$-6.5\,\mathrm{MHz}$	$-5.5\,\mathrm{MHz}$
$\delta_0/2\pi = +17.5\,\mathrm{MHz}$	99.944	99.870	99.519
$\delta_0/2\pi = +18.5\,\mathrm{MHz}$	99.984	**99.991**	99.958
$\delta_0/2\pi = +19.5\,\mathrm{MHz}$	99.801	99.918	99.833

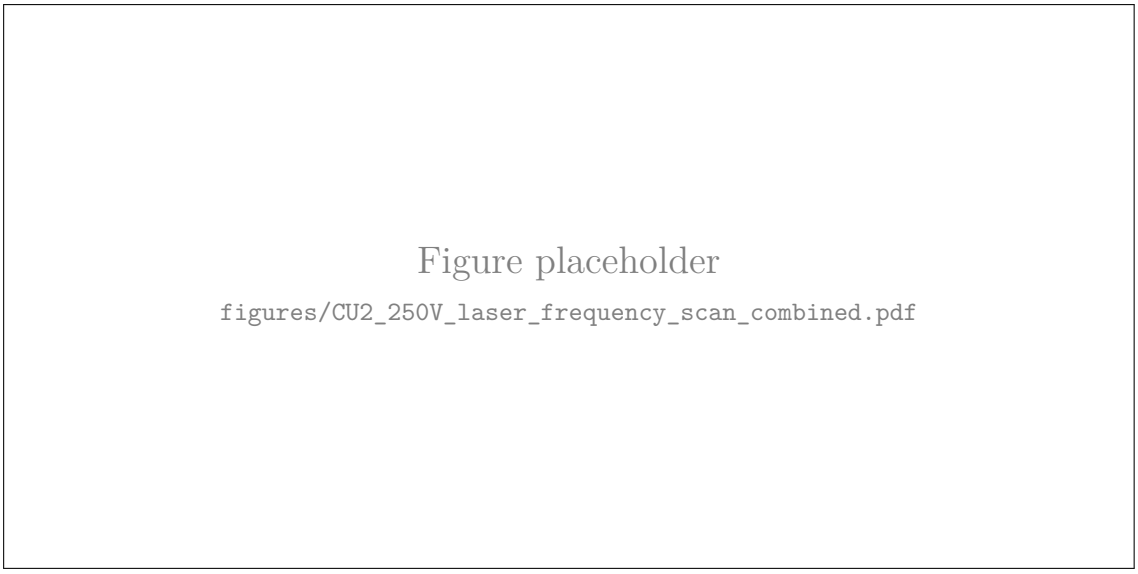


Figure B.1: CU2 at 250 V/cm: cleanup versus δ_0 at fixed $f_{\mathrm{AOM}} = 194.93\,\mathrm{MHz}$ (run `correlated_detuning_shared_offset_boarder_combined`). Reference only.

Single-transition variant. A variant with only R(2) $F' = 3$, in which one laser is split by an AOM into two phase-modulated tones (80 % and 20 % of 100 mW) offset by Δ_{AOM}, was simulated at 250 V/cm with a 38-level model. At $\delta/2\pi = \Delta_{\mathrm{AOM}}/2\pi = 18.5$ MHz it reaches $\eta_{\mathrm{c}} = 99.05\,\%$ with $N_\gamma = 1.53$, with the removed molecules in $J = 3$ (59.9 %), $J = 4$ (37.2 %) and $J = 5$ (1.9 %). It corresponds to step 2 of the design (Sec. 3.3), which the design study found limited by dark states, and is not pursued further.

Appendix C

Code and data

A typical run consists of

1. `lindblad.generate_OBE_system_transitions` (build the OBE system);
2. binding of the runtime parameters and `lindblad.prepare_lindblad_problem` (prepare once);
3. `lindblad.solve_lindblad` for single trajectories, or `grid_scan_with_progress` and `lindblad.parameter_scan` for scans;
4. `save_scan_hdf` for storage.

Stored runs are read back with `load_scan_hdf` without rebuilding the model.

Table C.1: Notebooks and scripts.

file	contents
`SPB_cleanup_J1-J2F2.ipynb`	CU1: build, single trajectory, detuning, power and geometry scans, HDF5 readback, optimum and re-run
`SPB_cleanup_J2_J3F4_opposite_parity.ipynb`	CU2 two-transition scheme: build, single trajectory, detuning scans, combination of runs
`SPB_cleanup_J2_J3F3_2laser.ipynb`	CU2 single-transition variant (250 V/cm; not pursued)
`scripts/scan_progress.py`	chunked grid scan with progress bar (`grid_scan_with_progress`)
`scripts/scan_hdf.py`	HDF5 storage with metadata (`save_scan_hdf`, `load_scan_hdf`)
`scripts/migrate_j2_j3f4_notebook.py`	one-off migration of the CU2 notebook from the Julia workflow
`scripts/refactor_scan_cells.py`	one-off rewrite of the scan sections
`scripts/cu2_laser_aom_scan.py`	CU2 two-dimensional scan in laser and AOM frequency at 200 V/cm: stages `test`, `coarse`, `fine`, `confirm`; checkpoints in `tmp/cu2_laser_aom_scan/`
`scripts/plot_cu2_laser_aom_map.py`	figures of the CU2 2D scan (Figs. 6.3 and 6.4)
`scripts/cu2_optimum_trajectories.py`	trajectories and photon numbers at the two CU2 optima, and their figures (Figs. 6.5 and 6.6)
`scripts/cu2_level_scheme.py`	CU2 excited-sublevel positions and coupling strengths at 200 V/cm, with the laser combs of the two optima (Fig. 6.7)
`scripts/cleanup_spb_doppler.py`	CU1 model with Doppler shift and optional EOM (`build_cu1`); SPB-related levels; (v_x, y_0) maps for CU1 and CU2
`scripts/plot_cu1_doppler_map.py`	Fig. 5.5
`scripts/cu1_eom_doppler_scan.py`	CU1 with the 1.6 MHz EOM: $\beta \times \delta \times v_x$ scan, y_0 averages, full map; checkpoints in `tmp/cu1_eom_doppler/`
`scripts/plot_cu1_eom.py`	Fig. 5.6
`scripts/cu1_trajectories.py`	CU1 trajectories per starting sublevel at +19.0 and +20.2 MHz (Fig. 5.7)
`scripts/cleanup_level_diagrams.py`	field-dressed level diagrams with laser lines and per-level residuals (slides only)
`scripts/cu1_3cm_scan.py`	CU1 with the 3 cm beam: detuning scan averaged over the acceptance, full map at the set point (Table 5.5, Fig. 5.8)
`scripts/cu2_doppler_scan.py`	CU2 (v_x, y_0) maps at R1 and R2, 3 cm beam, two initial states (Table 6.7)
`scripts/plot_cleanup_maps_current.py`	Figs. 5.8 and 6.8; mean in acceptance with the beam-simulation weights

Table C.2: Stored scan runs used in this report (best cleanup in the run). Both files are in `results/`; `results/scan_results.md` lists every run with its axes and conditions and is regenerated by `scripts/write_results_index.py`.

run	content	best η_c
`results/SPB_J1_cleanup_scans.h5` (CU1, 150 V/cm)		
`detuning_withoutP-_-20to_100MHz`	21 levels; $-20 \ldots 100$ MHz, 1 MHz	99.9275 %
`detuning_board_-20to_100MHz`	26 levels; $-20 \ldots 100$ MHz, 1 MHz	99.9269 %
`detuning_fine`	26 levels; $19 \ldots 22$ MHz, 0.1 MHz	99.963 %
`power_detuning_no_phase_eom`	26 levels; 10–110 mW × $-16 \ldots 30$ MHz	99.955 %
`vertical_position_beam_diameter`	earlier configuration; $y_0 \times 4\sigma_y$ at constant peak intensity	99.877 %
`doppler_cu1_spb`	26 levels; +20.2 MHz; v_x $-1 \ldots 1$ m/s × y_0 0–15 mm; SPB-related start	99.970 %
`doppler_cu1_uniform`	as above, uniform 12-level start	99.968 %
`doppler_cu1_spb_eom1p6MHz_beta1p5_det20MHz`	as `doppler_cu1_spb`, 1.6 MHz EOM, $\beta = 1.5$, +20.0 MHz	99.49 %
`doppler_cu1_spb_3cm_det19MHz`	3 cm beam, +19.0 MHz, no EOM; $v_x \times y_0$, 0.1 m/s × 1.5 mm	99.99 %
`eom_beta_detuning_vx_cu1_spb_4p8cm`	1.6 MHz EOM: $\beta \times \delta \times v_x$ at $y_0 = 0$ (NaN: not computed; Fig. 5.6a)	–
`doppler_cu1_spb_4p8cm_coarse_settings`	coarse $v_x \times y_0$ for the 9 settings of Table 5.4	–
`detuning_vx_y_cu1_spb_3cm_coarse`	3 cm beam: $\delta \times v_x \times y_0$ (Table 5.5)	–
`trajectories_cu1_vx0_19p0_20p2MHz_4p8cm`	populations vs time per starting sublevel (Fig. 5.7)	–
`per_level_cu1_20p2MHz_4p8cm`	remaining per starting sublevel, on axis	–
`per_state_cu1_spb_20p2MHz_4p8cm`	the 8 SPB-related levels, with destinations	–
`results/SPB_J2_cleanup_scans.h5` (CU2, 56 levels)		
`correlated_detuning_shared_offset_200V`	200 V/cm; $f_{\mathrm{AOM}} =$ 186.83 MHz: offset	99.33 %

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